

LOGIC AND TOPOLOGY ASPECTS OF THE PROBLEM OF HIDDEN VARIABLES IN QUANTUM MECHANICS

PRELIMINARY NOTES PREPARED BY J. IOVINO

For the University of Toronto–UT San Antonio Mathematics research group

ABSTRACT. We examine two independent proofs of the Bell–Kochen–Specker theorem on the impossibility of non-contextual hidden variables in quantum mechanics: Bell’s proof, which relies on Gleason’s theorem and elementary topology (and is thus infinitary in nature), and Kochen and Specker’s proof, which is finitary and constructive. Both arguments date from the mid-1960s. We show that, although they developed independently, they are intimately linked by a logical-topological fact: the compactness theorem of propositional logic. Along with the mathematical arguments, we provide detailed chronological timeline of historical developments around the problem of hidden variables.

0. PRELIMINARIES (ALL THE LINEAR ALGEBRA AND PHYSICS YOU WILL NEED TO READ THESE NOTES)

We assume that the reader is familiar with Hilbert spaces, with eigenvalues and eigenvectors of linear operators, and with orthogonal projections. Everything else we need, from both linear algebra and physics, is recalled in this short section.

Throughout these notes, H will denote a complex Hilbert space with $3 \leq \dim H \leq \infty$; everything below holds verbatim for real and quaternionic Hilbert spaces as well.

We use the physicists’ notation for vectors and functionals, which we now explain, since it is not standard in mathematics. The inner product $\langle \xi | \eta \rangle$ of $\xi, \eta \in H$ is linear in the second variable and conjugate-linear in the first. A vector η is also written $|\eta\rangle$ (a *ket*), and $\langle \xi |$ (a *bra*) denotes the linear functional $\eta \mapsto \langle \xi | \eta \rangle$. Juxtaposition is composition: for a bounded operator A , the number $\langle \xi | A | \eta \rangle$ means $\langle \xi | A \eta \rangle$, and $|\xi\rangle\langle \xi |$ is the operator $\eta \mapsto \langle \xi | \eta \rangle \xi$. We write $S(H)$ for the unit sphere of H .

Let $L(H)$ be the lattice of closed subspaces of H — an *orthomodular lattice* in the usual sense — which we identify with the corresponding orthogonal projections. Recall that the orthogonal projection P onto a closed subspace V is defined by writing each $\eta \in H$ uniquely as a sum $\eta = \eta_V + \eta_\perp$ with $\eta_V \in V$ and $\eta_\perp \in \perp V$, and putting $P\eta = \eta_V$. Two properties follow at once: first, $P^2 = P$, because P fixes every vector of V ; second, $P^* = P$ because for $\xi, \eta \in H$,

$$\langle \xi | P \eta \rangle = \langle \xi_V | \eta_V \rangle = \langle P \xi | \eta \rangle,$$

since the cross terms $\langle \xi_\perp | \eta_V \rangle$ and $\langle \xi_V | \eta_\perp \rangle$ vanish. Conversely, a bounded operator P with $P^2 = P = P^*$ is the orthogonal projection onto its range, which is closed. Thus, the elements of $L(H)$ are exactly the bounded operators P with $P^2 = P = P^*$, and from these two identities

$$\langle \eta | P | \eta \rangle = \langle \eta | P^* P | \eta \rangle = \langle P \eta | P \eta \rangle = \|P \eta\|^2 \quad (\eta \in H). \quad (1)$$

The lattice $L(H)$ carries a partial order $P \leq Q$ (inclusion of ranges), a meet $P \wedge Q$ (intersection of ranges), a join $P \vee Q$ (closed span of the ranges), an orthocomplement $P \mapsto P^\perp = I - P$, and extremal elements 0 and $I = H$. Two projections are *orthogonal*, $P \perp Q$, when their ranges are orthogonal, equivalently $P \leq Q^\perp$. A *basis* of $L(H)$ is the set of rank-one projections onto the vectors of some orthonormal basis of H .

For $\psi \in S(H)$, the operator $P_\psi = |\psi\rangle\langle\psi|$ is the orthogonal projection onto the one-dimensional subspace $\mathbb{C}\psi$: it fixes ψ , it kills every vector orthogonal to ψ , and $P_\psi^2 = P_\psi = P_\psi^*$. The rank-one projections are exactly the minimal nonzero elements, the *atoms*, of the lattice $L(H)$.

Now the physical meaning. In quantum mechanics, a projection $P \in L(H)$ represents a yes/no question about a physical system, that is, an experiment with two possible outcomes, and $P^\perp = I - P$ represents the same question with the answers exchanged. A state of the system assigns to each projection P the probability $\mu(P)$ of obtaining the answer “yes” to P . The question I is always answered “yes”, so $\mu(I) = 1$. If $P_0, P_1, \dots$ are pairwise orthogonal, the corresponding questions are mutually exclusive and can be tested together, and the probability of “yes” to their join $\bigvee_i P_i$ is the sum of the probabilities $\mu(P_i)$. These requirements are the axioms of a quantum measure (see Definition 1.1 below). Which assignments μ actually occur in physics is decided by the Born rule, which leads to the notion of density operator.



Remark 0.1 (The Born rule). The simplest states of a physical system, the *pure states*, are described by unit vectors ψ in the unit sphere $S(H)$ of H . The *Born rule* says that in the state ψ the probability of the answer “yes” to the question P is

$$\mu(P) = \|P\psi\|^2 (= \langle\psi|P|\psi\rangle, \text{ by (1)})$$

An important special case is the question “is the system in the state ξ ?” for another unit vector $\xi \in S(H)$. It is represented by the rank-one projection $P_\xi = |\xi\rangle\langle\xi|$ onto the line $\mathbb{C}\xi$, which maps a vector η to $\langle\xi|\eta\rangle\xi$. Since $P_\xi\psi = \langle\xi|\psi\rangle\xi$ and $\|\xi\| = 1$, the Born rule gives

$$\mu(P_\xi) = \|P_\xi\psi\|^2 = |\langle\xi|\psi\rangle|^2,$$

the familiar probability that a system in the state ψ is found in the state ξ .

A pure state represents complete knowledge of the system: no further information about how the system was prepared could sharpen the probabilities $\|P\psi\|^2$. Even so, these probabilities are in general strictly between 0 and 1. The randomness they describe is therefore not due to any lack of information on our part; according to quantum mechanics it is intrinsic to the system itself, and the theorems of these notes are precisely about whether it could be explained away as ignorance of some hidden variables.

In practice, however, even this complete knowledge is rarely available. Real physical sources do not produce a single pure state: the light of an ordinary lamp, for instance, consists of photons whose polarizations vary at random from one photon to the next, so no single vector describes the beam. Our information about how a system was prepared is usually incomplete, and the imperfections of the preparing apparatus mean that we know at best which pure states may have been produced, and with what probabilities. This introduces a second kind of randomness, of an entirely different nature from the first: it is classical randomness, of the same kind as in a game of cards, and it reflects only our ignorance, that is, the imperfection of our information about the system, not any property of the system itself. A state that combines the two kinds of randomness is called a *mixed state*, or a *mixture*.¹

Accordingly, suppose one knows only that the system has been prepared in the state ψ_k with probability λ_k , for a family of states $\psi_k \in S(H)$ and weights $\lambda_k \geq 0$ with $\sum_k \lambda_k = 1$. Here the weights λ_k express our ignorance of the preparation, while each $\langle\psi_k|P|\psi_k\rangle$ is a

¹The same mathematical description arises in a situation where ignorance is not the explanation: a system that has interacted with another one, and is therefore entangled with it, has in general no pure state of its own, even when the combined system is in a pure state, and its statistics are again those of a mixture.

quantum probability given by the Born rule. Combining the two, the probability of “yes” to P is then the weighted average (or expected value)

$$\sum_k \lambda_k \langle \psi_k | P | \psi_k \rangle. \quad (2)$$

As we shall see next, this expression depends on the preparation only through the operator

$$\rho = \sum_k \lambda_k |\psi_k\rangle\langle\psi_k| = \sum_k \lambda_k P_{\psi_k},$$

and it is natural to look for a single formula that computes (2) from ρ and P . The trace provides it:

When H is finite-dimensional, the *trace* of a linear operator T on H is

$$\text{Tr}(T) = \sum_i \langle e_i | T e_i \rangle,$$

where (e_i) is an orthonormal basis of H . It is the sum of the diagonal entries of the matrix of T in that basis, and it does not depend on the basis. When H is infinite-dimensional, the same formula defines the trace of the *trace-class* operators, for which the sum converges absolutely and is again independent of the basis. We shall only need two facts about them: every operator of finite rank, in particular every rank-one projection P_ψ , is trace-class; and if T is trace-class and A is bounded, then TA and AT are trace-class. For the definition and these facts, see [RS80, Section VI.6]. The reader who prefers may assume that H is finite-dimensional throughout this section; the infinite-dimensional case differs only in these points of analysis.

The following observation will play an important role in these notes.

Observation 0.2. *Let A be a bounded operator on H and let $\psi \in S(H)$. Then*

$$\text{Tr}(AP_\psi) = \text{Tr}(P_\psi A) = \langle \psi | A | \psi \rangle.$$

In particular, $\text{Tr}(P_\psi P) = \langle \psi | P | \psi \rangle = \|P\psi\|^2$ for every $P \in L(H)$, and $\text{Tr}(P_\xi P_\psi) = |\langle \xi | \psi \rangle|^2$ for $\xi \in S(H)$.

Proof. Compute the trace in an orthonormal basis (e_i) of H with $e_0 = \psi$. Then $P_\psi e_0 = \psi$, and $P_\psi e_i = 0$ for $i \geq 1$ because e_i is orthogonal to ψ . So only the first term survives:

$$\text{Tr}(AP_\psi) = \sum_i \langle e_i | AP_\psi e_i \rangle = \langle \psi | A \psi \rangle = \langle \psi | A | \psi \rangle,$$

Similarly, since $P_\psi A e_i = \langle \psi | A e_i \rangle \psi$ and $\langle e_i | \psi \rangle = 0$ for $i \geq 1$,

$$\text{Tr}(P_\psi A) = \sum_i \langle e_i | P_\psi A e_i \rangle = \sum_i \langle e_i | \psi \rangle \langle \psi | A e_i \rangle = \langle \psi | A \psi \rangle.$$

If $P \in L(H)$, then by (1) $\langle \psi | P \psi \rangle = \langle P \psi | P \psi \rangle = \|P\psi\|^2$. Finally $P_\psi \xi = \langle \psi | \xi \rangle \psi$, so taking $A = P_\xi$ gives $\text{Tr}(P_\xi P_\psi) = \|P_\psi \xi\|^2 = |\langle \xi | \psi \rangle|^2$. 

Returning to the Born rule: by Observation 0.2 and linearity of the trace, the probability (2) is

$$\sum_k \lambda_k \langle \psi_k | P | \psi_k \rangle = \sum_k \lambda_k \text{Tr}(P_{\psi_k} P) = \text{Tr}(\rho P).$$

So the Born rule, for pure and mixed preparations alike, takes the single form $P \mapsto \text{Tr}(\rho P)$. The operator ρ has three properties that do not depend on how it was obtained: it is positive, since $\langle \xi | \rho | \xi \rangle = \sum_k \lambda_k |\langle \psi_k | \xi \rangle|^2 \geq 0$; it is trace-class; and it has trace 1. The last property is

nothing but the requirement that the weights λ_k add up to 1: each P_{ψ_k} has trace $\text{Tr}(P_{\psi_k}) = \langle \psi_k | \psi_k \rangle = 1$ (take $A = I$ in Observation 0.2), so

$$\text{Tr}(\rho) = \sum_k \lambda_k \text{Tr}(P_{\psi_k}) = \sum_k \lambda_k = 1.$$

Equivalently, $\text{Tr}(\rho) = \text{Tr}(\rho I)$ is the probability of the answer “yes” to the question I , which is always answered “yes”. These are the properties singled out in the following definition.

Definition 0.3 (Density operator). A *density operator* (or *density matrix*) on H is a trace-class operator ρ that is positive, i.e. $\langle \xi | \rho | \xi \rangle \geq 0$ for all $\xi \in H$, and has $\text{Tr}(\rho) = 1$.

Conversely, every density operator arises from a preparation as in Remark 0.1. Suppose first that H is finite-dimensional. A positive operator ρ on a complex Hilbert space is self-adjoint, so ρ has an orthonormal basis of eigenvectors ψ_k ; the corresponding eigenvalues $\lambda_k = \langle \psi_k | \rho | \psi_k \rangle$ are nonnegative because ρ is positive, and they add up to $\text{Tr}(\rho) = 1$. Hence ρ can be written as

$$\rho = \sum_k \lambda_k |\psi_k\rangle\langle\psi_k|, \quad \lambda_k \geq 0, \quad \sum_k \lambda_k = 1, \quad (3)$$

where ψ_k has eigenvalue λ_k . When H is infinite-dimensional, the same decomposition holds with a countable orthonormal family (ψ_k) of eigenvectors, by the spectral theorem for compact self-adjoint operators [RS80, Section VI.5]; density operators are compact. So density operators are exactly the probabilistic mixtures of the *pure states* P_ψ , $\psi \in S(H)$.

Finally, taking $A = \rho$ in Observation 0.2, the probability of finding a system in the state ρ in the pure state ψ is

$$\text{Tr}(\rho P_\psi) = \langle \psi | \rho | \psi \rangle.$$

If ψ is an eigenvector of ρ with eigenvalue λ , since $\|\psi\| = 1$, the preceding scalar value must be λ ; in general it is a weighted average of the eigenvalues of ρ , namely $\sum_k \lambda_k |\langle \psi_k | \psi \rangle|^2$.

This is it, dear reader. This is all the quantum mechanics and all the linear algebra you will need for the rest of these notes. You now have somewhat less at your disposal than von Neumann had in 1932, and considerably more than he used in his proof that hidden variables are impossible.

1. QUANTUM MEASURES AND GLEASON’S THEOREM

Definition 1.1 (Quantum measure). A *quantum measure* (or *state*²) on $L(H)$ is a function $\mu : L(H) \rightarrow [0, 1]$ with $\mu(I) = 1$ that is countably additive on orthogonal families: for pairwise orthogonal $\{P_i\}$,

$$\mu\left(\bigvee_i P_i\right) = \sum_i \mu(P_i).$$

A *two-valued quantum measure* is a quantum measure $\mu : L(H) \rightarrow \{0, 1\}$.

Although it is motivated by physics (see Section 0), this definition is purely measure-theoretic, on an abstract orthomodular lattice; nothing in it refers to physics. (When $L(H)$ is read as the projection lattice of a physical system, a two-valued quantum measure is usually called a *valuation*; see Appendix A.1.)

²Gleason called this object a *measure* [Gle57]; the quantum-logic tradition (Mackey [Mac63], Piron [Pir64]) relabeled it a *state*, and much of the older literature still uses that name; later authors (e.g. [Ham03]) reverted to *quantum measure*, the term used throughout these notes.

Theorem 1.2 (Main Theorem, Bell–Kochen–Specker). *If $3 \leq \dim H \leq \infty$, no two-valued quantum measure exists on $L(H)$.*

We give two independent proofs of this: a topological one, due to Bell (see Section 3 below; it is the simpler of the two) and a combinatorial one, due to Kochen and Specker (see Section 5 below). Bell’s proof relies on Gleason’s theorem, which describes all quantum measures in terms of density operators (see Definition 0.3). For every density operator ρ , the map $P \mapsto \text{Tr}(\rho P)$ is a quantum measure: it takes values in $[0, 1]$, it gives $\text{Tr}(\rho) = 1$ at I , and it is additive on finite orthogonal families, because the join of finitely many pairwise orthogonal projections is their sum and the trace is linear. In finite dimension every orthogonal family of nonzero projections is finite, so this is all that is needed. In infinite dimension, countable additivity follows in the same way from the continuity of $A \mapsto \text{Tr}(\rho A)$ on bounded increasing sequences of projections [RS80, Section VI.6]. By Remark 0.1, these are exactly the quantum measures given by the Born rule. Gleason’s Theorem says that, conversely, every quantum measure arises in this way: in dimension at least 3, the Born rule is forced by the lattice structure of $L(H)$ alone.

We can now state Gleason’s theorem.

Theorem 1.3 (Gleason’s Theorem [Gle57]). *Let $3 \leq \dim H \leq \infty$. Every quantum measure μ on $L(H)$ has the form $\mu(P) = \text{Tr}(\rho P)$ for a unique density operator ρ (see Definition 0.3).*

We take this theorem as given, as its proof (Gleason’s original argument, or any later streamlining) is a substantial piece of real analysis. The following remark, which comes from Gleason’s paper [Gle57, p. 886], shows that the lower bound $\dim H \geq 3$ cannot be weakened to $\dim H \geq 2$. The remark is not needed for our proofs, so it can be skipped without loss of continuity.



Remark 1.4. We show that the lower bound $\dim H \geq 3$ is optimal. On $H = \mathbb{R}^2$, parametrize rays by angle $\theta \in [0, \pi)$. Every orthonormal basis is a pair $\{\theta, \theta + \pi/2 \pmod{\pi}\}$. Define a function $v : [0, \pi) \rightarrow \{0, 1\}$ by

$$v(\theta) = \begin{cases} 1, & \text{if } \theta \in [0, \pi/2), \\ 0, & \text{if } \theta \in [\pi/2, \pi). \end{cases}$$

For every such pair, exactly one of $\theta, \theta + \pi/2 \pmod{\pi}$ lands in $[0, \pi/2)$, so v satisfies additivity on every basis: v is a genuine quantum measure in the sense of Definition 1.1, two-valued, on $L(\mathbb{R}^2)$. Its associated function $g(\theta) := v(\theta)$ is visibly discontinuous (jumps at $\theta = 0, \pi/2$), so no density operator ρ satisfies $g(\theta) = \langle \theta | \rho | \theta \rangle$: v is a direct counterexample to the conclusion of Gleason’s Theorem at $\dim H = 2$.

2. THE “ONLY ONE WINNER PER BASIS” LEMMA

The following elementary lemma plays a pivotal role in these notes.

Lemma 2.1. *Let v be a two-valued quantum measure on $L(H)$ and $\{e_i\}_{i \in I}$ an orthonormal basis of H . Exactly one index $i \in I$ has $v(P_{e_i}) = 1$; the rest have $v(P_{e_i}) = 0$.*

Proof. For $i \neq j$ in I , orthogonality gives $v(P_{e_i}) + v(P_{e_j}) = v(P_{e_i} \vee P_{e_j}) \in \{0, 1\}$, so at most one index has value 1. Summing over the basis, $\sum_i v(P_{e_i}) = v(I) = 1$; since each term is 0 or 1, exactly one equals 1. 😊



Remarks 2.2.

- (1) The proof of Lemma 2.1 does not use $\dim H \geq 3$; it holds for any orthonormal basis of cardinality at least 2, in any dimension. This dimension-independence might suggest the lemma already settles Theorem 1.2 on its own; it does not.
- (2) The lemma only pins down, separately, one privileged ray in *each* orthonormal basis; it says nothing about how the privileged ray of one basis relates to that of a different, overlapping basis. The argument presented in the next section supplies exactly that missing ingredient — Gleason’s classification of *all* quantum measures, plus connectedness of $S(H)$ — to force the private choice of every basis to cohere into one single global function, and then breaks that global function against itself.³

3. BELL’S 1964 PROOF OF THEOREM 1.2, BASED ON GLEASON’S THEOREM AND ELEMENTARY TOPOLOGY

In these notes, we provide two proofs of the non-existence of hidden variables (see Theorem 1.2). In this section, we present Bell’s proof, which is based on Gleason’s Theorem and elementary topology, and in Section 5, we present the Kochen–Specker proof. Bell’s proof and the Kochen–Specker proof date from the mid-1960s but are independent of each other. In the Timeline section (see Section 6 below) we give a detailed list of historical developments.

Proof of Theorem 1.2, via Gleason’s Theorem and connectedness. Suppose v is a two-valued quantum measure on $L(H)$. By Gleason’s Theorem, $v(P) = \text{Tr}(\rho P)$ for a fixed density operator ρ , so $g(\psi) := v(P_\psi) = \langle \psi | \rho | \psi \rangle$ (see Observation 0.2) is continuous on $S(H)$ (a matrix element of the fixed bounded operator ρ) and $\{0, 1\}$ -valued. Since $S(H)$ is connected, a continuous $\{0, 1\}$ -valued function on it is constant: $g \equiv c$ for a fixed $c \in \{0, 1\}$. But by Lemma 2.1, on any orthonormal basis g equals 1 at exactly one point and 0 at the rest — it takes *both* values there, contradicting $g \equiv c$. ☺



Remark 3.1. The hypothesis $\dim H \geq 3$ is only used to invoke Gleason’s Theorem. Nothing else in the proof depends on dimension.



Remark 3.2 (Leonardo’s *sfumato*: quantum measures need a *sfumatura*). Bell’s proof can be read as a statement about painting, in the spirit of Leonardo da Vinci⁴. A two-valued quantum measure would be a picture of the sphere $S(H)$ is like a digital photo, built from 0’s and 1’s, with both digits appearing on every orthonormal basis (see Lemma 2.1), and hence with sharp transitions between colors. Gleason’s Theorem forbids sharp transitions. Every quantum measure has the form $\psi \mapsto \langle \psi | \rho | \psi \rangle$, a continuous function on the path-connected space $S(H)$, so it cannot pass from 0 to 1 without taking every value in between: if $\mu(P_u) = 1$ and $\mu(P_w) = 0$, then along any path from u to w in $S(H)$ the function $\psi \mapsto \mu(P_\psi)$ takes every value in $[0, 1]$, by the intermediate value theorem. For the pure state $\rho = |u\rangle\langle u|$ this *sfumatura* is the Born rule itself.

³We observe that $S(H)$ is connected; in fact, it is even path-connected: If $v \neq -u$, the path $t \mapsto \frac{(1-t)u+tv}{\|(1-t)u+tv\|}$, $t \in [0, 1]$, joins u to v inside $S(H)$, and $\theta \mapsto e^{i\theta}u$, $\theta \in [0, \pi]$, joins u to $-u$.

⁴Leonardo da Vinci’s *sfumato* (from the Italian *fumo*, smoke) is the technique of modeling light and shade through imperceptible gradations, with no hard contour lines; a phrase often attributed to Leonardo describes it as blending “without lines or borders, in the manner of smoke.” The soft transitions around the eyes and the corners of the mouth in Leonardo’s *Mona Lisa* are the classic example.

✂ **Historical context of Bell’s result.** Credit here divides among three researchers, and it is worth being precise about the division, since the history that follows can otherwise read as though it points in two directions at once. Gleason proved Gleason’s Theorem in 1957, with no stated connection to hidden variables. Jauch played two distinct roles. In 1963 he and his student Piron published a no-hidden-variables argument of their own [JP63], formulated in the lattice-theoretic setting used throughout these notes; and in the same year he pointed out to Bell that Gleason’s already-published Theorem bears on the hidden-variables question. The proof given in this section — the argument that combines Gleason’s Theorem with connectedness, or, equivalently, the value-clash argument of the Provenance paragraph just after this one, to derive Theorem 1.2 — was constructed by Bell, whose paper takes the Jauch–Piron argument as one of its targets. That division is why this section is titled for Bell alone: Gleason’s Theorem enters as a named, cited ingredient, and Jauch’s roles, as author of a rival argument and as the person who directed Bell to Gleason, belong to the history of how Bell came to write the proof, not to the authorship of the proof.

In more detail: in 1932 von Neumann gave the first no-hidden-variables argument, in *Mathematische Grundlagen der Quantenmechanik* [vN32]: it required any dispersion-free (hidden-variable) assignment to be linear even across non-commuting observables, an assumption criticized as physically unmotivated first by Grete Hermann in 1935 [Her35], whose critique went largely unread by physicists until Jammer drew attention to it [Jam74], and later, independently, by Bell: there is no operational reason a hidden-variable value assigned to $A + B$ should equal the sum of the values assigned to A and B when A, B do not commute and so cannot be jointly measured (see the Timeline section, Section 6, for Bell’s own, more colorful, verdict on this point). Gleason’s 1957 Theorem [Gle57] was motivated by a question of G. Mackey’s about measures on the closed subspaces of a Hilbert space [Mac57, Mac63], again with no stated bearing on the hidden-variables debate. The connection to hidden variables was drawn six years later: J. M. Jauch (1914–1974), then directing the Institute of Theoretical Physics at the University of Geneva, pointed out Gleason’s Theorem to Bell in 1963. At that time, Jauch was supervising Constantin Piron’s doctoral thesis at Geneva, *Axiomatique quantique* [Pir64], which recasts the propositional structure of quantum mechanics as an orthomodular lattice satisfying further axioms (the same setting used throughout these notes). Gleason’s Theorem, a classification of measures on exactly such a lattice, sat squarely inside Jauch’s research program; its bearing on von Neumann’s argument would have been immediately apparent to him. In the same year Jauch and Piron published their own argument, *Can Hidden Variables be Excluded in Quantum Mechanics?* [JP63]. Its key assumption is that if two propositions each have probability 1, so does their meet. For compatible propositions this is harmless; Bell’s objection [Bel66] is that for incompatible P and Q the meet $P \wedge Q$ cannot be tested by measuring P and Q , so there is no physical reason to require the property of dispersion-free states — the same kind of objection he raised against von Neumann’s additivity. Bell later recalled, specifically, that Jauch pointed out that Gleason’s Theorem gives a strengthening of von Neumann’s conclusion, with the additivity hypothesis required only for commuting (orthogonal) observables rather than for all observables [Bel87] — exactly the distinction between a *quantum measure* and a full lattice homomorphism. Bell records the debt in a footnote of [Bel66]: “I am much indebted to Professor Jauch for drawing my attention to this work.” Bell’s own professional appointment at CERN was in particle and accelerator physics; quantum foundations was, by his own account and that of his colleagues, pursued alongside that work rather than as part of it [Ber14]. He wrote up the present consequence in 1964 while on a leave of absence from CERN at the Stanford Linear Accelerator Center (SLAC) — a sabbatical that freed him to work on foundational questions

directly [Ber14] — and after a two-year publication delay it appeared as *On the Problem of Hidden Variables in Quantum Mechanics*, Rev. Mod. Phys. **38** (1966), 447–452 [Bel66]. (The Timeline section, Section 6, gives the fuller chronology. Kochen and Specker’s work came in two stages: papers presented in 1963 and 1964 and published in 1965 [KS65b, KS65a], which introduce partial Boolean algebras, and the 1967 paper on hidden variables [KS67]. Bell’s 1966 paper appeared between the two and is independent of both.)



Provenance of the argument. The preceding argument is, in its essential content, due to Bell. His own argument did not use the language of continuity or connectedness: he showed directly, using an auxiliary third orthogonal direction (available exactly because $\dim H \geq 3$), that two rays with v -value 1 and 0 cannot be arbitrarily close; since both values must occur on every basis (see Lemma 2.1), two such close, oppositely valued rays are unavoidable, a contradiction. What we have included here is the now-standard topological restatement of Bell’s argument.

4. OBTAINING A FINITE OBSTRUCTION VIA LOGICAL COMPACTNESS

Bell’s proof of Theorem 1.2 given in Section 3 proves non-existence over the whole (infinite, uncountable) sphere without ever exhibiting a finite culprit. The main observation of this section (rooted in mathematical logic rather than physics) shows that a finite culprit was *guaranteed* to exist the moment Bell’s proof (see Section 3) was known. The main result of the section is Corollary 4.2.

Fix $\dim H = d < \infty$. Let $V = S(H)/\sim$ be the set of rays (unit vectors modulo global phase factors, i.e., $u \sim e^{i\theta}u$ for $\theta \in \mathbb{R}$), and let $\mathcal{E}$ be the set of all orthonormal bases of H , each basis being a set of exactly d rays. Treat each ray $u \in V$ as a propositional symbol σ_u , and to each basis $e = \{u_1, \dots, u_d\} \in \mathcal{E}$ associate the propositional formula

$$\varphi_e := (\sigma_{u_1} \vee \dots \vee \sigma_{u_d}) \wedge \bigwedge_{1 \leq i < j \leq d} \neg(\sigma_{u_i} \wedge \sigma_{u_j}) \quad (\text{“exactly one } \sigma_{u_i} \text{ is true”}),$$

a formula in the d propositional symbols of e alone. A two-valued quantum measure on $L(H)$ is exactly a truth assignment to $\{\sigma_u : u \in V\}$ satisfying every φ_e simultaneously (see Lemma 2.1: “exactly one per basis” is the correct, and only, translation of the quantum-measure axioms into this propositional language).

Proposition 4.1. *If every finite subset of bases $\mathcal{E}_0 \subseteq \mathcal{E}$ admits some truth assignment to the (finitely many) rays occurring in $\mathcal{E}_0$ satisfying φ_e for every $e \in \mathcal{E}_0$, then a two-valued quantum measure exists on all of $L(H)$.*

Proof. This proposition is an instance of the Compactness Theorem for propositional logic (equivalently, compactness of the product space $\{0,1\}^V$ in the product topology, i.e. Tychonoff’s theorem for the two-point discrete space; see e.g. [End01]): a set T of propositional formulas, each in finitely many of a (possibly infinite or uncountable) set of propositional symbols, is simultaneously satisfiable if and only if every finite subset of T is. Apply this to $T = \{\varphi_e : e \in \mathcal{E}\}$; each φ_e involves only the d propositional symbols of its own basis e , so a finite $T_0 \subseteq T$ corresponds exactly to a finite $\mathcal{E}_0 \subseteq \mathcal{E}$, and satisfiability of T_0 is exactly satisfiability of $\{\varphi_e : e \in \mathcal{E}_0\}$ by some assignment to the finitely many rays involved. ☺

Corollary 4.2. *For $\dim H \geq 3$, there exists a finite set of rays and a finite set of orthonormal bases drawn from them admitting no assignment satisfying “exactly one per basis” on every one of those bases.*

Proof. Theorem 1.2 (via Section 3, and hence via Gleason’s theorem) says no two-valued quantum measure exists on $L(H)$, i.e. T is not satisfiable. By Proposition 4.1 (that is, by compactness) some finite $T_0 \subseteq T$ is not satisfiable, i.e. some finite set of bases $\mathcal{E}_0$, drawn from finitely many rays, admits no valid assignment. ☺



Remark 4.3. Corollary 4.2 is exactly “a finite Kochen–Specker set exists,” derived purely from Section 3 without constructing one. This derivation is genuinely non-constructive: the Compactness Theorem, proved via an ultrafilter/Tychonoff argument, uses (a weak form of) the Axiom of Choice, and gives no bound on the size or location of the finite witness.

The combinatorial proof given in the next section (see Section 5 below), in contrast, hands you an explicit culprit of size 18 with no appeal to the Axiom of Choice at all. The two finite-flavored facts — “a finite witness exists” and “here is an explicit one, consisting of 18 vectors” — are logically different achievements: the first is a two-paragraph corollary of connectedness plus a compactness theorem from 1930s logic; the second required constructing an explicit combinatorial object, which took from 1967 (Kochen–Specker, 117 vectors) to 1996 (18 vectors; see Section 5 below) to achieve economically, with the question of the true minimum still open in dimension 3 (see the [2024](#) entry of the Timeline section, Section 6).

5. KOCHEN AND SPECKER’S 1967 PROOF OF THEOREM 1.2, BASED ON A CONCRETE FINITE OBSTRUCTION

The proof of Theorem 1.2 given in this section requires no measure theory and no continuity. It replaces Bell’s use, in Section 3, of Gleason and connectedness with a finite, checkable substitute supplying the same missing ingredient: a way to force the separate “one winner per basis” choices of Lemma 2.1 to interact.

We work in $H = \mathbb{R}^4$. Gleason’s Theorem and Theorem 1.2 hold there by the same proofs as for $\mathbb{C}^4$.

Lemma 5.1 (Cabello–Estebaranz–García-Alcaine [CEGA96]). *The following 9 orthonormal bases of $\mathbb{R}^4$ (each a list of 4 pairwise orthogonal vectors) are built from 18 distinct vectors in total, each of which occurs in exactly 2 of the 9 bases.*

$$\begin{aligned}
 B_1 &= \{(0, 0, 0, 1), (0, 0, 1, 0), (1, 1, 0, 0), (1, -1, 0, 0)\} \\
 B_2 &= \{(0, 0, 0, 1), (0, 1, 0, 0), (1, 0, 1, 0), (1, 0, -1, 0)\} \\
 B_3 &= \{(1, -1, 1, -1), (1, -1, -1, 1), (1, 1, 0, 0), (0, 0, 1, 1)\} \\
 B_4 &= \{(1, -1, 1, -1), (1, 1, 1, 1), (1, 0, -1, 0), (0, 1, 0, -1)\} \\
 B_5 &= \{(0, 0, 1, 0), (0, 1, 0, 0), (1, 0, 0, 1), (1, 0, 0, -1)\} \\
 B_6 &= \{(1, -1, -1, 1), (1, 1, 1, 1), (1, 0, 0, -1), (0, 1, -1, 0)\} \\
 B_7 &= \{(1, 1, -1, 1), (1, 1, 1, -1), (1, -1, 0, 0), (0, 0, 1, 1)\} \\
 B_8 &= \{(1, 1, -1, 1), (-1, 1, 1, 1), (1, 0, 1, 0), (0, 1, 0, -1)\} \\
 B_9 &= \{(1, 1, 1, -1), (-1, 1, 1, 1), (1, 0, 0, 1), (0, 1, -1, 0)\}.
 \end{aligned}$$

Verification. Orthogonality within each B_k is a finite check of $\binom{4}{2} = 6$ dot products, e.g. for B_3 : $(1, -1, 1, -1) \cdot (1, -1, -1, 1) = 1 + 1 - 1 - 1 = 0$, $(1, -1, 1, -1) \cdot (1, 1, 0, 0) = 0$, $(1, -1, 1, -1) \cdot (0, 0, 1, 1) = 0$, $(1, -1, -1, 1) \cdot (1, 1, 0, 0) = 0$, $(1, -1, -1, 1) \cdot (0, 0, 1, 1) = 0$, $(1, 1, 0, 0) \cdot (0, 0, 1, 1) = 0$; the other eight bases check the same way. For the incidence count, tabulate the occurrences of each vector: $(0, 0, 0, 1) \in B_1, B_2$; $(0, 0, 1, 0) \in B_1, B_5$; $(1, 1, 0, 0) \in B_1, B_3$; $(1, -1, 0, 0) \in B_1, B_7$; $(0, 1, 0, 0) \in B_2, B_5$; $(1, 0, 1, 0) \in B_2, B_8$; $(1, 0, -1, 0) \in B_2, B_4$;

$(1, -1, 1, -1) \in B_3, B_4$; $(1, -1, -1, 1) \in B_3, B_6$; $(0, 0, 1, 1) \in B_3, B_7$; $(1, 1, 1, 1) \in B_4, B_6$; $(0, 1, 0, -1) \in B_4, B_8$; $(1, 0, 0, 1) \in B_5, B_9$; $(1, 0, 0, -1) \in B_5, B_6$; $(0, 1, -1, 0) \in B_6, B_9$; $(1, 1, -1, 1) \in B_7, B_8$; $(1, 1, 1, -1) \in B_7, B_9$; $(-1, 1, 1, 1) \in B_8, B_9$. Each of the 18 vectors occurs in exactly 2 bases, and $9 \text{ bases} \times 4 \text{ vectors} = 36 = 18 \times 2$, confirming the count is exhaustive. 

Proof of Theorem 1.2, combinatorial, after Kochen and Specker [KS67], in the streamlined form of [CEGA96]. Suppose a two-valued quantum measure v exists on $L(\mathbb{R}^4)$. Restrict attention to the 18 rays of Lemma 5.1. Since each B_k is an orthonormal basis, additivity forces

$$\sum_{u \in B_k} v(u) = v(I) = 1 \quad \text{for each } k = 1, \dots, 9.$$

Summing over all 9 bases,

$$\sum_{k=1}^9 \sum_{u \in B_k} v(u) = 9.$$

Reorganize the left side by vector instead of by basis: each of the 18 vectors u occurs in exactly 2 of the B_k , so it contributes $2v(u)$ to the double sum. Hence,

$$\sum_{k=1}^9 \sum_{u \in B_k} v(u) = \sum_u 2v(u) = 2 \sum_u v(u),$$

which is an even integer. However, the same quantity was computed above to equal 9, an odd integer, which is an immediate contradiction. 

 **Historical context of the Kochen–Specker result.** Kochen and Specker published *The Problem of Hidden Variables in Quantum Mechanics* [KS67] in 1967, giving an independent, self-contained combinatorial proof using 117 vectors in $\mathbb{R}^3$ (see Figure 1 below), phrased in the language of *partial Boolean algebras*: a model-theoretic framework treating the compatible (commuting) observables at each context as an ordinary Boolean algebra, glued together across contexts, and asking whether this partial structure embeds in a single total Boolean algebra, i.e. admits a two-valued homomorphism. Both authors came to the problem as logicians rather than physicists, which accounts for the difference in idiom from Bell’s route through Gleason’s Theorem (see Section 3). Ernst Specker (1920–2011), professor at ETH Zürich from 1955, worked across mathematical logic, set theory, and combinatorics — among other results, he had disproved the axiom of choice in Quine’s *New Foundations* a decade earlier — and had already announced the present obstruction, without proof, in 1960 [Spe60], saying only that “an elementary geometric argument” (*ein elementargeometrisches Argument*) establishes it. Simon Kochen (b. 1934) wrote his 1958 Princeton doctoral thesis, under Alonzo Church, on ultraproducts and arithmetical extensions of models, and in these same years was completing, with James Ax, the papers on valued fields that became the Ax–Kochen theorem; he joined the Princeton mathematics faculty in 1967, the year this paper appeared. This common background as logicians — model theory specifically for Kochen, logic, set theory, and combinatorics for Specker — is what makes the partial-Boolean-algebra embedding formulation the natural way for these two authors to have posed the question, rather than Bell’s measure-theoretic route through Gleason’s Theorem. (The Timeline section, Section 6, gives the fuller chronology.) The proof given above is not Kochen and Specker’s original 117-vector construction but Cabello, Estebaranz, and García-Alcaine’s 1996 streamlining of it to 18 vectors [CEGA96]. The [1991](#), [1994](#), and [1996](#) entries of the Timeline section (see

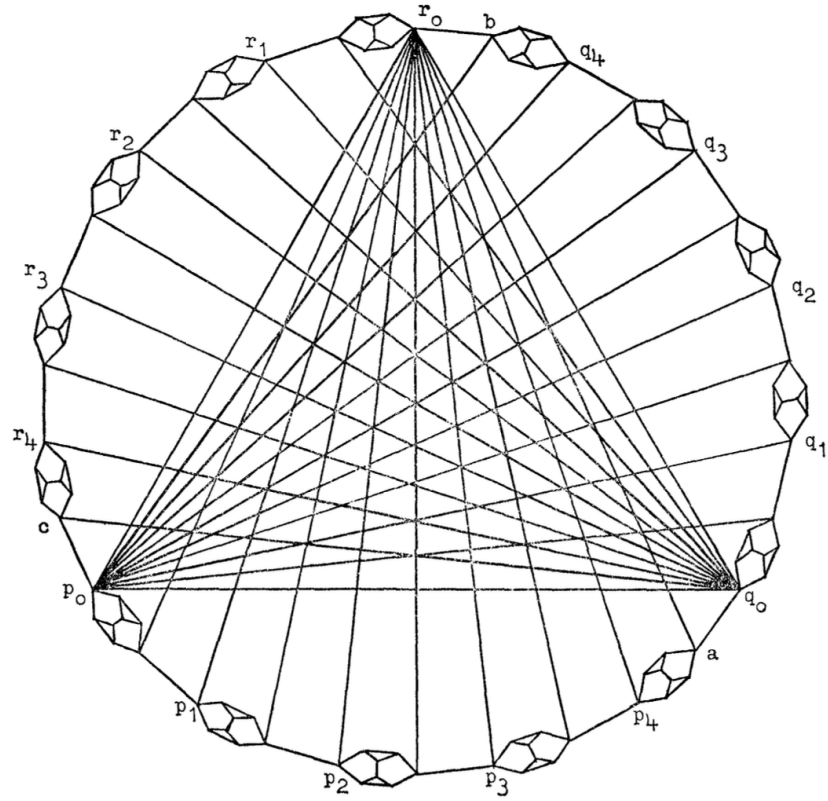
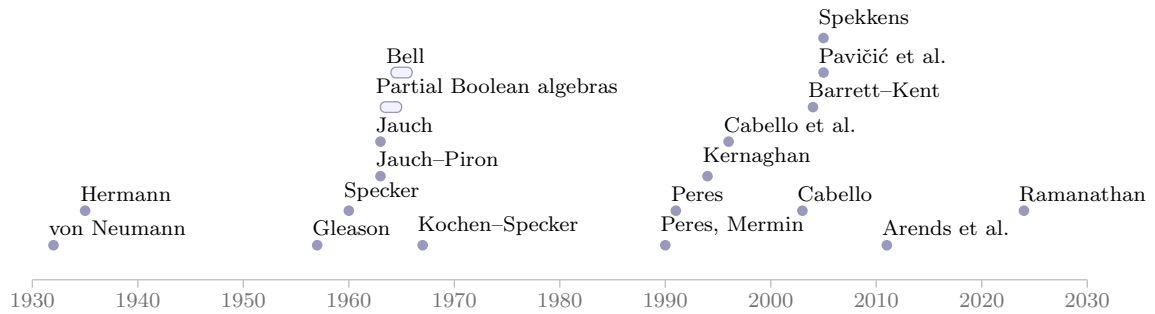


FIGURE 1. The graph Γ_2 of Kochen and Specker's 1967 paper [KS67, p. 69], whose 117 vertices are the vectors of their original proof; two vertices are joined by an edge when the corresponding vectors are orthogonal. The vertices are the points on the rim, after identifying p_0 with a , q_0 with b , and r_0 with c ; each of the fifteen small gadgets on the rim is a copy of their ten-vertex graph Γ_1 .

Section 6 below) record how that reduction came about, by way of intermediate constructions of Peres and Kernaghan.

6. TIMELINE



- 1932 • **von Neumann.** Von Neumann gives the first no-hidden-variables argument, in *Mathematische Grundlagen der Quantenmechanik* [vN32]; see Section 3 for the argument itself and its later criticism by Bell. Bell put that criticism more colorfully in a later interview: “the proof of von Neumann is not merely false but foolish!” [MC88].
- 1935 • **Hermann.** Grete Hermann, in *Die naturphilosophischen Grundlagen der Quantenmechanik* [Her35], objects that von Neumann’s proof assumes, for hypothetical dispersion-free states, an additivity of expectation values justified only for quantum states. Her critique went largely unnoticed by physicists until the 1970s [Jam74]; how closely it matches Bell’s later objection is still debated [MS18].
- 1957 • **Gleason.** Gleason proves Gleason’s Theorem [Gle57], motivated by a question of G. Mackey’s about measures on the closed subspaces of a Hilbert space [Mac57, Mac63], with no stated connection to the hidden-variables debate; see Section 3.
- 1960 • **Specker.** In *Die Logik nicht gleichzeitig entscheidbarer Aussagen* [Spe60], Specker asserts that the closed subspaces of a three-dimensional Hilbert space cannot be embedded in a Boolean algebra compatibly with the operations on orthogonal elements, saying only that “an elementary geometric argument” (*ein elementargeometrisches Argument*) shows this; however, no proof is given.
- 1963 • **Jauch–Piron.** Jauch and Piron publish *Can Hidden Variables be Excluded in Quantum Mechanics?* [JP63], a no-hidden-variables argument in the lattice-theoretic setting, resting on the assumption that two propositions of probability 1 have a meet of probability 1. Bell’s 1966 paper criticizes this assumption for incompatible propositions; see Section 3.
- 1963 • **Jauch.** J. M. Jauch (1914–1974), a Swiss-American theoretical physicist then directing the Institute of Theoretical Physics at the University of Geneva, points out Gleason’s Theorem to Bell as relevant to the hidden-variables question, at the same time as he is supervising Piron’s doctoral thesis on a lattice-theoretic reformulation of quantum mechanics [Pir64]. See Section 3 for the fuller context, including Bell’s own account of what Jauch specifically pointed out [Bel87] and Bell’s acknowledgment of him in a footnote of [Bel66].
- 1963–1965 • **Partial Boolean algebras.** Kochen and Specker present *Logical Structures Arising in Quantum Theory* at the 1963 Berkeley symposium on the theory of models and *The Calculus of Partial Propositional Functions* at the 1964 Jerusalem congress; both appear in 1965 [KS65b, KS65a]. These papers introduce partial Boolean algebras; the application to hidden variables would come later, in their 1967 paper.

- 1964–1966

Bell. Bell writes *On the Problem of Hidden Variables in Quantum Mechanics* [Bel66], submitted in 1964 but delayed in publication for two years, until Rev. Mod. Phys. **38** (1966), 447. It criticizes von Neumann’s 1932 argument and the 1963 Jauch–Piron argument, and gives an elementary route — close in spirit to Section 3 — from Gleason’s Theorem to the non-existence of non-contextual hidden variables, deliberately avoiding the full strength of Gleason’s proof. Bell’s paper thus appeared after Kochen and Specker’s papers on partial Boolean algebras (see the 1963–1965 entry) but before their 1967 paper on hidden variables, and was written independently of both.
- 1967

Kochen–Specker. Kochen and Specker publish *The Problem of Hidden Variables in Quantum Mechanics* [KS67], an independent, self-contained combinatorial proof using 117 vectors in $\mathbb{R}^3$, phrased in the language of partial Boolean algebras; see Section 5 for the logical idiom of that proof and for Kochen’s and Specker’s own backgrounds as logicians.
- 1990

Peres, Mermin. Peres and, independently and shortly after, Mermin introduce the “magic square” proof: a 3×3 grid of two-qubit Pauli observables whose row and column products force an algebraic contradiction for any non-contextual ± 1 assignment. It proves a related but formally different statement (state-independent contextuality for an algebra of observables, rather than the ray-coloring statement of Theorem 1.2) and is, like the proof of Section 5, a finite parity argument.
- 1991

Peres. Peres reduces Kochen and Specker’s construction to 33 vectors in $\mathbb{R}^3$, and gives a 24-vector construction in $\mathbb{R}^4$ [Per91]. Conway and Kochen later find 31 vectors in $\mathbb{R}^3$ (unpublished; reported in [Per93]).
- 1994

Kernaghan. Kernaghan finds 20-vector Kochen–Specker sets in $\mathbb{R}^4$ inside Peres’s 24-vector configuration [Ker94].
- 1996

Cabello et al. Cabello, Estebaranz, and García-Alcaine give the 18-vector construction in $\mathbb{R}^4$ of Section 5 [CEGA96]. Their construction did not start from scratch: it builds on Peres’s 24-vector configuration in $\mathbb{R}^4$ (see the 1991 entry) and on the 96 distinct 20-vector critical subsets identified within it (see the 1994 entry). Cabello, Estebaranz, and García-Alcaine found that certain 18-vector subsets are themselves critical, and noted, in their own words, that none of these is contained in any of the 96 known 20-vector critical sets, “which probably explains why they were not obtained previously.”
- 2003

Cabello. Cabello extends contextuality to POVMs for a single qubit [Cab03] (see Appendix B below).
- 2004

Barrett–Kent. Barrett and Kent identify the finite-precision loophole for fuzzy, probabilistic assignments [BK04] (see Appendix B below).

- 2005

Pavičić et al. Pavičić, Merlet, McKay, and Megill show, by exhaustive computer enumeration of Kochen–Specker sets built from complete orthonormal bases, that 18 is the minimum number of vectors in dimension 4 [PMMM05]: the Cabello–Estebaranz–García-Alcaine configuration, and hence the proof of Section 5, is not merely economical but optimal.
- 2005

Spekkens. Spekkens gives the operational definition of generalized non-contextuality closing that loophole [Spe05] (see Appendix B below).
- 2011

Arends et al. Arends, Ouaknine, and Wampler prove, by computer-assisted search, that any Kochen–Specker vector system in dimension 3 must contain at least 18 vectors [AOW11], the first of the lower bounds recorded in the 2024 entry.
- 2024

Ramanathan. Ramanathan generalizes the outcome alphabet from $\{0, 1\}$ towards $[0, 1]$ [Ram24] (see Appendix B below). In parallel, the analogous minimality question in dimension 3 — open since Kochen and Specker’s original 117-vector construction — remains unsettled: the best known explicit set has 31 vectors (an unpublished construction of Conway and Kochen, reported in [Per93]), while the best proved lower bound, obtained by SAT-solver and computer-algebra methods, stands at 24 [LBG24].

6.1. Why Kochen and Specker, and not Gleason, became the famous result. Given Bell’s proof of Section 3, it is fair to ask why the timeline just given runs through 1967 at all: the non-existence of a two-valued quantum measure follows in a few lines from Gleason’s 1957 Theorem, so why is the 1967 paper, and not the 1957 one, the result every student of quantum foundations learns by name? A few distinct things are responsible, and none of them is that the later authors found a technically deeper fact.

First, Gleason’s own paper contains no trace of the hidden-variables question. It was written to settle a question of Mackey’s about measures on the closed subspaces of a Hilbert space, with no stated bearing on non-contextuality, dispersion-free states, or the EPR debate. Nothing in it flags the two-valued case as distinguished. The consequence for hidden variables is visible to us because the valuation formalism connecting it to them (see Appendix A.1 below) is already set up; in 1957 that connection had not been drawn by anyone, Gleason included. It took until 1963 (see the 1963 entry of the Timeline) for anyone to notice it could be used this way — six years in which the fact sat in print doing no work on the question it is now most famous for answering.

Second, once noticed, Bell’s proof had bad luck getting into print (see the 1964–1966 entry of the Timeline): the two-year publication delay meant his paper appeared only in 1966, just a year before Kochen and Specker’s 1967 paper on hidden variables, which became the standard reference.

Third, and most importantly, Kochen and Specker’s proof is not a corollary of anything. It is self-contained: finite, combinatorial, checkable by hand, and it borrows nothing from measure theory or continuity. This self-containedness is a real asset independent of priority. A reader deciding whether non-contextual hidden variables are possible can verify the combinatorial proof of Section 5 completely without trusting an external classification theorem about the whole continuum of rays of a Hilbert space — Gleason’s proof, by contrast, is substantial analysis, read by a comparatively narrow specialist audience even today. Being finite and elementary also made the result generative in a way a measure-theoretic corollary

is not: it invited thirty years of shrinking the witness (from 117 to 33 and then 31 vectors in $\mathbb{R}^3$, and from 24 to 20 to 18 in $\mathbb{R}^4$, the last since proved optimal [PMMM05]), it generalizes to settings — POVMs, abstract hypergraphs of contexts, generalized probabilistic theories — with no Hilbert-space trace structure for Gleason’s Theorem to even apply to, and it describes a finite, fixed set of compatible measurements that can be, and has been, physically implemented in a laboratory, which “all quantum measures on $L(H)$ at once” cannot.

So the historical asymmetry is not that Kochen and Specker found something Gleason missed; by the letter of the mathematics, he had it first. It is that recognizing the significance of a fact for a specific open question, and then re-deriving it in the most economical and self-contained form for the audience already concerned with that question, is itself a distinct and evidently more consequential contribution than being the first to establish the fact in some other, unrelated context. The same pattern recurs with Bell’s own inequality theorem, which eclipsed any more abstract precursor for exactly the same reasons: elementary, checkable, and aimed directly at the question the field already cared about.

7. QUANTUM MEASURES HAVE NO COUNTERPART OF DIRAC MEASURES AND DO NOT FORM A LATTICE

In this section we show that two basic properties of classical probability have no counterparts for quantum measures when $\dim H \geq 3$: (C1) a Dirac measure turns intersections of events into minima of values, and (C2) the finite measures on a σ -algebra form a lattice (we prove this in Proposition 7.1 below). We first state (C1) and (C2) precisely. We then formulate their quantum counterparts as two questions, (Q1) and (Q2), state the answers, and reformulate them. The rest of the section contains the proofs. Throughout this section, H is a Hilbert space with $\dim H \geq 2$.

The classical properties (C1) and (C2). In classical probability, the events form a σ -algebra Σ of subsets of a sample space Ω .

- (C1) The Boolean operations on events can be read off from the values of the Dirac measures: for every $\omega \in \Omega$ and all $A, B \in \Sigma$,

$$\delta_\omega(A \cap B) = \min(\delta_\omega(A), \delta_\omega(B)), \quad \delta_\omega(\Omega \setminus A) = 1 - \delta_\omega(A). \quad (4)$$

(Each side of each equation equals 1 if ω belongs to the event on the left-hand side, and 0 otherwise.)

- (C2) Let $\mathcal{M}(\Sigma)$ be the set of finite positive measures on Σ , and order $\mathcal{M}(\Sigma)$ by $\mu \leq \nu$ if $\mu(A) \leq \nu(A)$ for every $A \in \Sigma$. Then $(\mathcal{M}(\Sigma), \leq)$ is a lattice (see Proposition 7.1 below).

The quantum questions (Q1) and (Q2). To show that (C1) and (C2) have no quantum counterparts, we must first say what a counterpart would be. In quantum mechanics, the role of the σ -algebra Σ is played by the lattice $L(H)$, and the role of the probability measures is played by the quantum measures (see Definition 1.1). For a density operator ρ we write $\mu_\rho(P) = \text{Tr}(\rho P)$. Question (Q1) below asks whether (C1) has a counterpart in this setting, and question (Q2) asks whether (C2) has one.

- (Q1) For a quantum measure μ on $L(H)$, the counterpart of the first equation in (4) is

$$\mu(P \wedge Q) = \min(\mu(P), \mu(Q)). \quad (5)$$

Is there a quantum measure μ on $L(H)$ that satisfies (5) whenever $P, Q \in L(H)$ commute?

We make two comments on this formulation. First, (Q1) does not need to ask for a counterpart of the second equation in (4): the equation $\mu(P^\perp) = 1 - \mu(P)$ holds for every quantum measure, by additivity. Second, (Q1) asks for (5) only for commuting pairs, because commuting projections lie in a common Boolean subalgebra of $L(H)$, and a Boolean algebra is the setting of the classical equations (4). (If P and Q commute, the projections PQ , $P(I - Q)$, $(I - P)Q$ and $(I - P)(I - Q)$ are pairwise orthogonal with sum I ; the joins of subsets of these four projections form a Boolean subalgebra of $L(H)$, and it contains $P = PQ + P(I - Q)$ and $Q = PQ + (I - P)Q$.) (For arbitrary pairs, see Proposition 7.3 below.)

- (Q2) Let $\mathcal{C}(H)$ be the set of functions $P \mapsto \text{Tr}(TP)$ on $L(H)$, where T ranges over the positive trace-class operators on H . (When $\dim H \geq 3$, Gleason's Theorem shows that $\mathcal{C}(H)$ is the set of nonnegative multiples of quantum measures.) As in (C2), we allow all nonnegative multiples of measures, not only the normalized ones, because a greatest lower bound of two probability measures is in general not a probability measure (for example, the greatest lower bound of two distinct Dirac measures is 0). Suppose that we order $\mathcal{C}(H)$ by

$$\mu \leq \nu \quad \text{if } \mu(P) \leq \nu(P) \text{ for every } P \in L(H).$$

Is $(\mathcal{C}(H), \leq)$ a lattice?

The main result of this section.

- When $\dim H \geq 3$, the answer to (Q1) is negative (see Theorem 7.7 below); when $\dim H = 2$, it is positive (see Remark 1.4).
- When $\dim H \geq 2$, the answer to (Q2) is negative (see Proposition 7.10 below).

Thus, when $\dim H \geq 3$, neither (C1) nor (C2) has a quantum counterpart.

Reformulations and consequences of the answers. Recall that an *order isomorphism* between partially ordered sets $(X, \leq)$ and $(Y, \leq)$ is a bijection $\Phi : X \rightarrow Y$ such that, for all $x, x' \in X$, $x \leq x'$ if and only if $\Phi(x) \leq \Phi(x')$.

- *The answer to (Q1), reformulated.* In Lemma 7.2 below, we show that a quantum measure satisfies (5) for all commuting P, Q if and only if it is two-valued. Hence, for $\dim H \geq 3$, the negative answer to (Q1) is equivalent to the statement that there is no two-valued quantum measure on $L(H)$. This is Theorem 1.2. (In the terminology of Section A.1, a two-valued quantum measure is a non-contextual hidden-variable assignment.) Moreover, when $\dim H \geq 2$, no measure μ_ρ given by a density operator satisfies (5) for all commuting P, Q (see Corollary 7.8 below).
- *A consequence of the answer to (Q2).* For every σ -algebra Σ on a set Ω , there is no order isomorphism between $(\mathcal{C}(H), \leq)$ and $(\mathcal{M}(\Sigma), \leq)$ (see Corollary 7.11 below). This consequence holds when $\dim H = 2$, where two-valued quantum measures exist (see Remark 1.4). Hence it cannot be derived from the nonexistence of two-valued quantum measures.

Plan of the proofs. The proofs do not use Theorem 1.2.

- In Section 7.1 we prove (C2). (Property (C1) needs no proof: it holds by the remark following (4).)
- In Section 7.2 we prove Lemma 7.2, and we show that if (5) is required for all pairs of projections, not only commuting ones, then no quantum measure satisfies it.

- In Section 7.3 we answer (Q1) for $\dim H \geq 3$. The argument is topological: we give $L(H)$ the norm topology and the quantum measures the topology of pointwise convergence, and we show that a two-valued quantum measure cannot be continuous on the connected space of rays (see Proposition 7.6 below), whereas Gleason's Theorem makes every quantum measure continuous there.
- In Section 7.4 we answer (Q2).
- In Section 7.5 we show, using the compactness argument of Section 4, that the negative answer to (Q1) is already forced by finitely many projections in a 3-dimensional subspace of H .

7.1. The classical case. In this subsection we prove (C2). We show that, for given $\mu, \nu \in \mathcal{M}(\Sigma)$, we can split Ω into a set where $\mu \geq \nu$ and a set where $\mu \leq \nu$; what makes this split possible is the Hahn decomposition theorem, which we recall below. We then define a measure that agrees with ν on the first set and with μ on the second, and in Proposition 7.1 below we show that it is the greatest lower bound of μ and ν .

A *finite signed measure* on Σ is a countably additive function $\lambda : \Sigma \rightarrow \mathbb{R}$, for example $\mu - \nu$. Given a finite signed measure λ , a *Hahn set* for λ is a set $S \in \Sigma$ such that $\lambda(B) \geq 0$ for every measurable $B \subseteq S$ and $\lambda(B) \leq 0$ for every measurable $B \subseteq \Omega \setminus S$. The *Hahn decomposition theorem* says that every finite signed measure has a Hahn set (see, e.g., [Rud87, Theorem 6.14]).

Given $\mu, \nu \in \mathcal{M}(\Sigma)$ and a Hahn set S for $\mu - \nu$, define measures $\gamma, \eta \in \mathcal{M}(\Sigma)$ by

$$\gamma(A) = \nu(A \cap S) + \mu(A \setminus S), \quad \eta(A) = \mu(A \cap S) + \nu(A \setminus S) \quad (A \in \Sigma). \quad (6)$$

Proposition 7.1. *Let $\mu, \nu \in \mathcal{M}(\Sigma)$, and let S be a Hahn set for $\mu - \nu$. Then the measures γ and η defined in (6) are, respectively, the greatest lower bound and the least upper bound of μ and ν in $(\mathcal{M}(\Sigma), \leq)$.*

Proof. Since S is a Hahn set for $\mu - \nu$, we have $\mu(B) \geq \nu(B)$ for every measurable $B \subseteq S$ and $\mu(B) \leq \nu(B)$ for every measurable $B \subseteq \Omega \setminus S$. Hence, for every $A \in \Sigma$, $\nu(A \cap S) \leq \mu(A \cap S)$ and $\mu(A \setminus S) \leq \nu(A \setminus S)$, so $\gamma(A) \leq \mu(A)$ and $\gamma(A) \leq \nu(A)$; that is, γ is a lower bound of μ and ν . If $\kappa \in \mathcal{M}(\Sigma)$ is another lower bound, then for every $A \in \Sigma$, $\kappa(A) = \kappa(A \cap S) + \kappa(A \setminus S) \leq \nu(A \cap S) + \mu(A \setminus S) = \gamma(A)$. The argument for η is the same with the inequalities reversed. $\odot$

7.2. Quantum measures that preserve commuting meets. In this subsection we show that the condition in (Q1) forces a quantum measure to take only the values 0 and 1, as a Dirac measure does; thus (Q1) asks exactly whether two-valued quantum measures exist (see Lemma 7.2 below). We also show that if the condition in (Q1) is required for all pairs of projections, not only commuting ones, then no quantum measure satisfies it when $\dim H \geq 2$ (see Proposition 7.3 below). To state these results conveniently, we say that a quantum measure μ on $L(H)$ *preserves commuting meets* if μ satisfies (5) whenever $P, Q \in L(H)$ commute.

Lemma 7.2. *A quantum measure on $L(H)$ preserves commuting meets if and only if it is two-valued.*

Proof. Let μ be a quantum measure. By additivity, $\mu(0) = 0$ and $\mu(P^\perp) = 1 - \mu(P)$ for every P . Suppose that μ preserves commuting meets. For every P , the projections P and $P^\perp$ commute and $P \wedge P^\perp = 0$, so (5) gives $\min(\mu(P), 1 - \mu(P)) = 0$, that is, $\mu(P) \in \{0, 1\}$.

Conversely, suppose that μ is two-valued. Two orthogonal projections cannot both have value 1, since their join would have value 2. Let $P, Q \in L(H)$ commute. For commuting projections, PQ is the projection onto the intersection of their ranges, so $R = P \wedge Q = PQ$;

similarly $P' = P \wedge Q^\perp = P(I - Q)$ and $Q' = P^\perp \wedge Q = (I - P)Q$. These three projections are pairwise orthogonal, with $P = R + P'$ and $Q = R + Q'$. If $\mu(R) = 1$, then $\mu(P) = \mu(Q) = 1$. If $\mu(R) = 0$, then $\mu(P) = \mu(P')$ and $\mu(Q) = \mu(Q')$, and these values are not both 1 because P' and Q' are orthogonal. In either case, $\mu(R) = \min(\mu(P), \mu(Q))$. ☺

The first paragraph of the proof also shows exactly where (5) fails for a quantum measure that is not two-valued: if μ is a quantum measure and $0 < \mu(P) < 1$ for some P , then (5) fails for the commuting pair $P, P^\perp$, since $\min(\mu(P), \mu(P^\perp)) > 0 = \mu(P \wedge P^\perp)$.

Question (Q1) asks for (5) only for commuting pairs. For a Dirac measure, the first equation in (4) holds for all pairs of events, so one may also ask whether a quantum measure can satisfy the equation in (5) for all pairs of projections. The answer to this stronger question is negative whenever $\dim H \geq 2$, and the proof is elementary.

Proposition 7.3. *No quantum measure μ on $L(H)$ satisfies $\mu(P \wedge Q) = \min(\mu(P), \mu(Q))$ for all $P, Q \in L(H)$.*

Proof. Suppose that μ is such a quantum measure. By Lemma 7.2, μ is two-valued. Summing over an orthonormal basis $\{e_i\}$ of H gives $\sum_i \mu(P_{e_i}) = 1$, so $\mu(P_e) = 1$ for some $e \in S(H)$. Let K be a 2-dimensional subspace containing e , and let P_K be the projection onto K . Since $P_K \geq P_e$, we have $\mu(P_K) = 1$. For every orthonormal basis $\{a, b\}$ of K , $P_a + P_b = P_K$, so exactly one of $\mu(P_a), \mu(P_b)$ equals 1. Fix an orthonormal basis $\{a, b\}$ of K and let $c = (a + b)/\sqrt{2}$ and $d = (a - b)/\sqrt{2}$, so that $\{c, d\}$ is an orthonormal basis of K with no ray in common with $\{a, b\}$. Interchanging a and b , and c and d , if necessary, we may assume that $\mu(P_a) = \mu(P_c) = 1$. Then $P_a \neq P_c$. Two distinct rays meet in 0, so $\mu(P_a \wedge P_c) = \mu(0) = 0$, whereas $\min(\mu(P_a), \mu(P_c)) = 1$. ☺

7.3. Topology and the negative answer to (Q1). In this subsection we prove that the answer to (Q1) is negative when $\dim H \geq 3$, and we explain the role that topology plays in this answer. By Lemma 7.2, the question is whether two-valued quantum measures exist. We put natural topologies on $L(H)$ and on the quantum measures, compare them, and then show that a two-valued quantum measure cannot be continuous, whereas Gleason's Theorem makes every quantum measure continuous when $\dim H \geq 3$.

Notation. As in Section 0, $P_\psi = |\psi\rangle\langle\psi|$ denotes the projection onto $\mathbb{C}\psi$ for $\psi \in S(H)$. We write $\mu_\psi = \mu_{P_\psi}$, so that $\mu_\psi(P) = \langle\psi|P|\psi\rangle$ (see Observation 0.2).

The topologies. We give $L(H)$ the operator-norm topology, $d(P, Q) = \|P - Q\|$. We give the quantum measures the topology of pointwise convergence on $L(H)$: a net μ_α converges to μ if and only if $\mu_\alpha(P) \rightarrow \mu(P)$ for every $P \in L(H)$. The following proposition collects the facts about these topologies that we need. Items (1), (2) and (4) show that the rank-one projections and the measures μ_ψ form homeomorphic connected spaces; item (3) identifies the topology of pointwise convergence with the trace-norm topology on density operators; item (5) is used in the proof of Proposition 7.9.

Proposition 7.4.

- (1) For $\psi, \varphi \in S(H)$, $\|P_\psi - P_\varphi\| = (1 - |\langle\psi|\varphi\rangle|^2)^{1/2}$ and $\|P_\psi - P_\varphi\|_1 = 2(1 - |\langle\psi|\varphi\rangle|^2)^{1/2}$.
- (2) The set $\{P_\psi : \psi \in S(H)\}$ of rank-one projections is a connected subset of $L(H)$, and $P \mapsto P^\perp$ is an isometry of $L(H)$.
- (3) Let $3 \leq \dim H < \infty$. The map $\rho \mapsto \mu_\rho$ is a homeomorphism from the density operators with the trace norm onto the quantum measures with the topology of pointwise convergence.
- (4) The map $P_\psi \mapsto \mu_\psi$ is a homeomorphism from the set of rank-one projections onto the set $\{\mu_\psi : \psi \in S(H)\}$. In particular, the latter set is connected.

(5) For density operators ρ, σ and $P, Q \in L(H)$, $|\mu_\rho(P) - \mu_\sigma(Q)| \leq \|\rho - \sigma\|_1 + \|P - Q\|$.

Proof. (1) Write $\lambda = (1 - |\langle \psi | \varphi \rangle|^2)^{1/2}$. The operator $X = P_\psi - P_\varphi$ is self-adjoint, has trace 0, and has range in $\text{span}(\psi, \varphi)$. So its nonzero eigenvalues are $\pm t$ for some $t \geq 0$, and, by Observation 0.2, $2t^2 = \text{Tr } X^2 = \text{Tr } P_\psi + \text{Tr } P_\varphi - 2\text{Tr}(P_\psi P_\varphi) = 2 - 2|\langle \psi | \varphi \rangle|^2$. Hence $t = \lambda$, $\|X\| = \lambda$ and $\|X\|_1 = 2\lambda$.

(2) By (1), $\|P_\psi - P_\varphi\| = (1 - |\langle \psi | \varphi \rangle|^2)^{1/2} \leq (2\|\psi - \varphi\|)^{1/2}$, because $1 - |\langle \psi | \varphi \rangle|^2 \leq 2(1 - \text{Re}\langle \psi | \varphi \rangle) = 2\text{Re}\langle \psi | \psi - \varphi \rangle \leq 2\|\psi - \varphi\|$. So $\psi \mapsto P_\psi$ is continuous, and the set of rank-one projections is the continuous image of the connected space $S(H)$. Finally, $P^\perp - Q^\perp = (I - P) - (I - Q) = Q - P$.

(3) By Gleason's Theorem, $\rho \mapsto \mu_\rho$ is a bijection onto the quantum measures. If $\|\rho_\alpha - \rho\|_1 \rightarrow 0$, then $|\mu_{\rho_\alpha}(P) - \mu_\rho(P)| = |\text{Tr}((\rho_\alpha - \rho)P)| \leq \|\rho_\alpha - \rho\|_1 \rightarrow 0$ for every P . Conversely, suppose that $\mu_{\rho_\alpha}(P) \rightarrow \mu_\rho(P)$ for every P . Fix an orthonormal basis $\{e_i\}$ of H , and for indices $i \neq j$ put $u = (e_i + e_j)/\sqrt{2}$ and $w = (e_i + ie_j)/\sqrt{2}$. For every density operator σ , expanding $\langle u | \sigma | u \rangle$ and $\langle w | \sigma | w \rangle$ gives

$$\text{Re}\langle e_i | \sigma | e_j \rangle = \langle u | \sigma | u \rangle - \frac{1}{2}\langle e_i | \sigma | e_i \rangle - \frac{1}{2}\langle e_j | \sigma | e_j \rangle, \quad \text{Im}\langle e_i | \sigma | e_j \rangle = \frac{1}{2}\langle e_i | \sigma | e_i \rangle + \frac{1}{2}\langle e_j | \sigma | e_j \rangle - \langle w | \sigma | w \rangle.$$

Each term on the right is μ_σ of a rank-one projection, so every matrix entry of ρ_α converges to the corresponding entry of ρ . In finite dimension this implies $\|\rho_\alpha - \rho\|_1 \rightarrow 0$.

(4) Let $\psi, \varphi \in S(H)$, and let (ψ_α) be a net in $S(H)$. The map is injective: if $\mu_\psi = \mu_\varphi$, then $|\langle \psi | \varphi \rangle|^2 = \mu_\psi(P_\varphi) = \mu_\varphi(P_\varphi) = 1$, so $P_\psi = P_\varphi$. If $\|P_{\psi_\alpha} - P_\psi\| \rightarrow 0$, then $\|P_{\psi_\alpha} - P_\psi\|_1 \rightarrow 0$ by (1), so $|\mu_{\psi_\alpha}(P) - \mu_\psi(P)| = |\text{Tr}((P_{\psi_\alpha} - P_\psi)P)| \leq \|P_{\psi_\alpha} - P_\psi\|_1 \rightarrow 0$ for every P . Conversely, if $\mu_{\psi_\alpha} \rightarrow \mu_\psi$ pointwise, then $|\langle \psi_\alpha | \psi \rangle|^2 = \mu_{\psi_\alpha}(P_\psi) \rightarrow \mu_\psi(P_\psi) = 1$, so $\|P_{\psi_\alpha} - P_\psi\| \rightarrow 0$ by (1). Connectedness follows from (2).

(5) $|\text{Tr}(\rho P) - \text{Tr}(\sigma Q)| \leq |\text{Tr}((\rho - \sigma)P)| + |\text{Tr}(\sigma(P - Q))| \leq \|\rho - \sigma\|_1 \|P\| + \|\sigma\|_1 \|P - Q\|$, and $\|P\| \leq 1, \|\sigma\|_1 = 1$. ☺

Comparison with the classical case. To see what goes wrong for $L(H)$, we compare the measures μ_ψ with their classical counterparts, the Dirac measures. Let Ω be a finite set with the discrete topology, and let Σ be the set of all subsets of Ω . For every $A \in \Sigma$, the function $\omega \mapsto \delta_\omega(A)$ takes only the values 0 and 1, and the set $\{\omega : \delta_\omega(A) = 1\} = A$ is both open and closed in Ω . The next proposition shows that both properties fail for the measures μ_ψ on $L(H)$.

Proposition 7.5. *Let $P \in L(H)$ with $P \neq 0, I$. The function $\psi \mapsto \mu_\psi(P)$ on $S(H)$ is continuous and takes every value in $[0, 1]$. The set $\{\psi \in S(H) : \mu_\psi(P) = 1\}$ is closed, nonempty, and not open.*

Proof. The function $\psi \mapsto \langle \psi | P | \psi \rangle$ is continuous; it equals 1 at the unit vectors in the range of P , and 0 at the unit vectors in the range of $P^\perp$. That it takes every value in $[0, 1]$ follows from the intermediate value theorem along a path in $S(H)$ from a unit vector in the range of P to one in the range of $P^\perp$, exactly as in Remark 3.2. The set where the function equals 1 is closed, nonempty and proper; if it were open, $S(H)$ would not be connected. ☺

Proposition 7.5 fixes a projection P and lets the measure μ_ψ vary. We now fix a measure and let the projection vary. A Dirac measure takes only the values 0 and 1; the quantum measures with this property are the two-valued ones. The next proposition shows that if v is such a measure, then the function $\psi \mapsto v(P_\psi)$ is not continuous.

Proposition 7.6. *If v is a two-valued quantum measure on $L(H)$, then $\psi \mapsto v(P_\psi)$ is not continuous on $S(H)$.*

Proof. Bell's proof of Theorem 1.2 in Section 3 uses Gleason's Theorem only to show that $\psi \mapsto v(P_\psi)$ is continuous on $S(H)$. If continuity is assumed instead, the rest of that proof applies verbatim. That part of the proof uses only Lemma 2.1 and the connectedness of $S(H)$, and therefore it does not require $\dim H \geq 3$. ☺

Combining Proposition 7.6 with Gleason's Theorem, which makes every quantum measure continuous on rays when $\dim H \geq 3$, we obtain the answer to (Q1).

Theorem 7.7. *A quantum measure on $L(H)$ that preserves commuting meets exists if $\dim H = 2$, and does not exist if $\dim H \geq 3$.*

Proof. Let $\dim H \geq 3$, and let μ be a quantum measure that preserves commuting meets. By Lemma 7.2, μ is two-valued. By Gleason's Theorem, $\mu = \mu_\rho$ for a density operator ρ , so $\psi \mapsto \mu(P_\psi) = \langle \psi | \rho | \psi \rangle$ is continuous on $S(H)$. This contradicts Proposition 7.6.

If $\dim H = 2$, then, as in Remark 1.4, assign the value 1 to one ray in each pair of orthogonal rays and the value 0 to the other, and set $v(0) = 0$ and $v(I) = 1$. This defines a two-valued quantum measure v , so v preserves commuting meets by Lemma 7.2. ☺

The measures μ_ρ given by density operators are the quantum measures used in physics (see Remark 0.1). Whatever the dimension of H , none of them satisfies the condition in (Q1).

Corollary 7.8. *For every density operator ρ , the quantum measure μ_ρ does not preserve commuting meets.*

Proof. If μ_ρ preserved commuting meets, it would be two-valued by Lemma 7.2. Since the function $\psi \mapsto \mu_\rho(P_\psi) = \langle \psi | \rho | \psi \rangle$ is continuous, this would contradict Proposition 7.6. ☺

In dimension 2, Gleason's Theorem fails (see Remark 1.4), and the answer to (Q1) is positive. The two-valued quantum measures that give the positive answer are, by Proposition 7.6, necessarily discontinuous on $S(H)$.

So far, the obstruction has come from the continuity of quantum measures. In the last result of this subsection we show that the lattice operations of $L(H)$ themselves are not continuous for the norm topology, and that no continuous rule computes $\mu_\rho(P \wedge Q)$ from $\mu_\rho(P)$ and $\mu_\rho(Q)$. By contrast, for a Dirac measure the rule is min, by (4), and min is continuous.

Proposition 7.9. *Let P_n and P be rank-one projections with $P_n \neq P$ and $P_n \rightarrow P$ in $L(H)$. Then $P_n \wedge P = 0$ and $\|P_n \vee P - P\| = 1$ for every n , while $P \wedge P = P \vee P = P$. Moreover, for every density operator ρ there is no continuous $m : [0, 1]^2 \rightarrow [0, 1]$ with $\mu_\rho(Q \wedge R) = m(\mu_\rho(Q), \mu_\rho(R))$ for all $Q, R \in L(H)$.*

Proof. Since $P_n \neq P$, the ranges of P_n and P are distinct lines, so $P_n \wedge P = 0$. The join $P_n \vee P$ is the projection onto a 2-dimensional space containing the range of P , so $P_n \vee P - P$ is a rank-one projection and has norm 1.

Suppose such an m exists, and let $P = P_\psi$ be any rank-one projection. Choose a unit vector $\varphi \perp \psi$ and put $P_n = P_{\psi_n}$ with $\psi_n = (\cos \frac{1}{n})\psi + (\sin \frac{1}{n})\varphi$. Then $P_n \neq P$ and $P_n \rightarrow P$ by Proposition 7.4(1), and $\mu_\rho(P_n) \rightarrow \mu_\rho(P)$ by Proposition 7.4(5). Hence

$$0 = \mu_\rho(P_n \wedge P) = m(\mu_\rho(P_n), \mu_\rho(P)) \rightarrow m(\mu_\rho(P), \mu_\rho(P)) = \mu_\rho(P \wedge P) = \mu_\rho(P).$$

So μ_ρ vanishes on every rank-one projection. Summing over an orthonormal basis gives $\mu_\rho(I) = 0$, a contradiction. ☺

7.4. The negative answer to (Q2). In this subsection we prove that the answer to (Q2) is negative whenever $\dim H \geq 2$ (see Proposition 7.10 below). We first translate the order on $\mathcal{C}(H)$ into the usual order on self-adjoint operators, so that (Q2) becomes a question about positive operators. Recall that $\mathcal{C}(H) = \{\mu_T : T \geq 0 \text{ trace class}\}$, where $\mu_T(P) = \text{Tr}(TP)$, and that $\mathcal{C}(H)$ is ordered pointwise on $L(H)$. For positive trace-class operators S and T , we have $\mu_S \leq \mu_T$ if and only if $S \leq T$. Indeed, if $\mu_S \leq \mu_T$, then $\langle \psi | T - S | \psi \rangle = \mu_T(P_\psi) - \mu_S(P_\psi) \geq 0$ for every $\psi \in S(H)$, so $T - S \geq 0$; and if $T - S \geq 0$, then $\text{Tr}((T - S)P) = \text{Tr}(P(T - S)P) \geq 0$ for every $P \in L(H)$, so $\mu_S \leq \mu_T$.

To show that $\mathcal{C}(H)$ is not a lattice, it suffices to exhibit two elements of $\mathcal{C}(H)$ without a greatest lower bound. We choose two commuting operators, so that the failure cannot be attributed to noncommutativity of the operators. For orthonormal $e_1, e_2 \in H$, let

$$A = P_{e_1} + P_{e_2}, \quad D = 2P_{e_1} + \frac{1}{2}P_{e_2}. \quad (7)$$

Proposition 7.10. *Let $e_1, e_2 \in H$ be orthonormal, and let A and D be the operators defined in (7). Then μ_A and μ_D have no greatest lower bound in $\mathcal{C}(H)$, although A and D commute.*

Proof. Let $K = \text{span}(e_1, e_2)$, and let μ_T be a lower bound of μ_A and μ_D in $\mathcal{C}(H)$, so that $0 \leq T \leq A$ and $T \leq D$. If $\xi \perp K$, then $0 \leq \langle \xi | T | \xi \rangle \leq \langle \xi | A | \xi \rangle = 0$, so $T^{1/2}\xi = 0$ and hence $T\xi = 0$. Since T is self-adjoint, it follows that T maps K into K and vanishes on $K^\perp$. We may therefore identify the lower bounds of μ_A and μ_D with positive 2×2 matrices in the basis e_1, e_2 . In this basis we have the first two of the following matrices; the other two will be used below:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \quad \tau = \begin{pmatrix} \frac{11}{12} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

The matrix M is a lower bound of A and D . Suppose that G were a greatest lower bound. Then $M \leq G \leq A$ forces $G_{11} = 1$. Since $G - M \geq 0$ has a zero $(1, 1)$ entry, its off-diagonal entries vanish (a positive 2×2 matrix with a zero diagonal entry has zero off-diagonal entries, because its determinant is nonnegative). Then $M \leq G \leq D$ forces $G_{22} = \frac{1}{2}$, so $G = M$. We now show that τ is a lower bound of A and D that is not below M ; this contradicts $G = M$. Recall that a self-adjoint 2×2 matrix is positive if and only if its diagonal entries and its determinant are nonnegative. Now $\tau \geq 0$ (its determinant is $\frac{1}{6}$), $A - \tau = \begin{pmatrix} \frac{1}{12} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} \end{pmatrix} \geq 0$ (its determinant is 0), and $D - \tau = \begin{pmatrix} \frac{13}{12} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} \end{pmatrix} \geq 0$ (its determinant is $\frac{5}{24}$), while $M - \tau = \begin{pmatrix} \frac{1}{12} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} \end{pmatrix}$ has determinant $-\frac{1}{24} < 0$, so $\tau \not\leq M$. Hence no greatest lower bound exists. 

The example shows where the classical argument breaks down. If we allowed only diagonal matrices, that is, operators that commute with P_{e_1} and P_{e_2} , then M would be the greatest lower bound of A and D , just as in the classical case (see Proposition 7.1): indeed, $\text{diag}(a, b) \leq A$ and $\text{diag}(a, b) \leq D$ if and only if $a \leq 1$ and $b \leq \frac{1}{2}$. The lower bound τ , which shows that M is not the greatest lower bound, is not diagonal.

Proposition 7.10 has the following consequence (see the definition of order isomorphism at the beginning of this section).

Corollary 7.11. *Let Σ be a σ -algebra on a set Ω . There is no order isomorphism between $(\mathcal{C}(H), \leq)$ and $(\mathcal{M}(\Sigma), \leq)$.*

Proof. Suppose that $\Phi : \mathcal{C}(H) \rightarrow \mathcal{M}(\Sigma)$ is an order isomorphism. Then Φ^{-1} is also an order isomorphism. Let $\mu, \nu \in \mathcal{C}(H)$. By (C2) (see Proposition 7.1), $\Phi(\mu)$ and $\Phi(\nu)$ have a greatest lower bound γ in $\mathcal{M}(\Sigma)$. We claim that $\Phi^{-1}(\gamma)$ is a greatest lower bound of μ and ν in $\mathcal{C}(H)$.

Since γ is a lower bound of $\Phi(\mu)$ and $\Phi(\nu)$, $\Phi^{-1}(\gamma)$ is a lower bound of μ and ν . If $\kappa \in \mathcal{C}(H)$ is a lower bound of μ and ν , then $\Phi(\kappa)$ is a lower bound of $\Phi(\mu)$ and $\Phi(\nu)$, so $\Phi(\kappa) \leq \gamma$, and hence $\kappa \leq \Phi^{-1}(\gamma)$. This proves the claim. Thus any two elements of $\mathcal{C}(H)$ have a greatest lower bound, which contradicts Proposition 7.10. 



Remark 7.12. We record two known theorems related to Proposition 7.10. They are not used in these notes. Recall that a compact convex subset of a finite-dimensional real vector space is a *simplex* if each of its points is a convex combination of its extreme points in exactly one way. A finite-dimensional real vector space ordered by a closed generating cone is a lattice if and only if a base of the cone is a simplex [Alf71]. For $H = \mathbb{C}^2$ the base of $\mathcal{C}(H)$ is the set of density operators, and it is not a simplex: $\frac{1}{2}I = \frac{1}{2}(P_\psi + P_{\psi'})$ for every orthonormal basis $\{\psi, \psi'\}$ of $\mathbb{C}^2$. Kadison [Kad51] proved that two self-adjoint operators on a Hilbert space have a least upper bound in the self-adjoint operators only if they are comparable.

7.5. A finite witness for the negative answer to (Q1). In this subsection we show that the negative answer to (Q1) is already forced by finitely many projections in a 3-dimensional subspace of H (see Theorem 7.13 below). This is of interest because the proof of Theorem 7.7 applies Gleason's Theorem to all of $L(H)$, an infinite object. We derive the finite statement from Bell's argument and the compactness argument of Section 4, both applied in $\mathbb{C}^3$, and then transfer it to every Hilbert space of dimension at least 3.

Theorem 7.13. *There is a finite family $\mathcal{E}_0$ of orthonormal bases of $\mathbb{C}^3$ such that no function from the rays occurring in $\mathcal{E}_0$ to $\{0, 1\}$ takes the value 1 at exactly one ray of each basis in $\mathcal{E}_0$. Consequently, if $3 \leq \dim H \leq \infty$, then no quantum measure on $L(H)$ preserves commuting meets.*

Proof. The finite family. This is Corollary 4.2 with $d = 3$. Notice that the proof of that corollary uses Theorem 1.2 only for $H = \mathbb{C}^3$, and only through Bell's argument in Section 3, which combines Gleason's Theorem in $\mathbb{C}^3$ with the connectedness of the unit sphere; it then applies Proposition 4.1. A unitary map carries $\mathcal{E}_0$ to a family with the same property, so $\mathcal{E}_0$ can be placed in any 3-dimensional space.

Transfer to H . Let v be a quantum measure on $L(H)$ that preserves commuting meets. By Lemma 7.2, v is two-valued. Summing over an orthonormal basis $\{e_i\}$ of H gives $\sum_i v(P_{e_i}) = 1$, so $v(P_e) = 1$ for some $e \in S(H)$. Let K be a 3-dimensional subspace containing e , and let P_K be the projection onto K . Since $P_e \leq P_K$, we have $v(P_K) = 1$. By the first step, we may assume that $\mathcal{E}_0$ consists of orthonormal bases of K . For each basis $\{u_1, u_2, u_3\}$ of K in $\mathcal{E}_0$, $P_{u_1} + P_{u_2} + P_{u_3} = P_K$, so $v(P_{u_1}) + v(P_{u_2}) + v(P_{u_3}) = 1$: exactly one of the three rays has value 1. Thus the restriction of v to the rays occurring in $\mathcal{E}_0$ takes the value 1 at exactly one ray of each basis in $\mathcal{E}_0$, which the first step rules out. This contradiction proves the theorem. 

The proof shows that the negative answer to (Q1) is witnessed by finitely many commuting families of three rank-one projections each, namely the images of the bases in $\mathcal{E}_0$ under a unitary map of $\mathbb{C}^3$ onto a single 3-dimensional subspace of H . Similarly, the witnesses in the proofs of Propositions 7.3 and 7.10 (the rays P_a, P_c , and the matrices M, τ) lie in a single 2-dimensional subspace of H .

APPENDIX A. PHYSICAL INTERPRETATION AND EXTENSIONS

Nothing in this appendix is a proof of Theorem 1.2. The appendix explains what the theorem is *about* physically, and why a purely mathematical uniqueness-of-classification fact

(Gleason) does not, by itself, close off the questions the theorem was designed to answer. Readers who are only interested in mathematics can skip this appendix.

A.1. What a two-valued quantum measure represents. When H is the state space of a quantum system, closed subspaces correspond to yes/no physical properties (a system either does or does not have its state in the range of P), and a quantum measure $\mu(P)$ is read as the probability that a measurement of the property P returns “yes.” A *two-valued* quantum measure v is then the mathematical shadow of a specific physical hypothesis: that every property has a definite, pre-existing truth value, the same value however and whenever it is measured, jointly consistent with the sum rule that mutually compatible (orthogonal) properties obey. This hypothesis is precisely what is meant by a *non-contextual hidden-variable assignment*. So Theorem 1.2, read physically, says: quantum mechanics (for any system with $\dim H \geq 3$) admits no such assignment. This physical reading is the sense in which the proof of Theorem 1.2 in Section 3 — which is nothing but Gleason’s Theorem plus Lemma 2.1 plus connectedness — already *is* a complete proof that Gleason’s Theorem rules out non-contextual hidden variables.

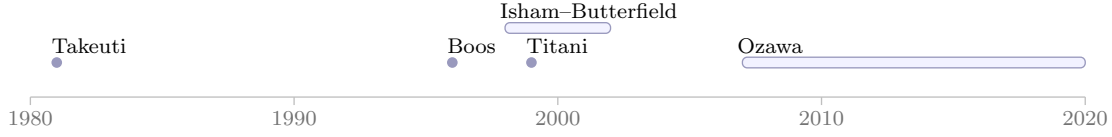
A.2. Why prove it again, physically. Section 4 and the discussion following the Timeline already gave the purely mathematical reasons (logical strength, constructivity) for the further proofs. Two more reasons live squarely in physics, not mathematics, and are worth stating as such rather than dressed up as further rigor:

- (1) *Experimental realizability.* The proof of Gleason’s Theorem concerns *all* quantum measures on the whole (continuum-sized) $L(H)$ at once; no laboratory ever measures “all of $L(H)$.” The 18 rays and 9 bases of Section 5, by contrast, are a finite, fixed list of compatible measurement settings that has actually been implemented experimentally to test contextuality. A physical no-go theorem that hands you a finite protocol is answering a different, additional question — “can this be tested?” — that a classification theorem about the whole lattice does not address.
- (2) *Portability beyond Hilbert space.* The statement “no consistent $\{0, 1\}$ -assignment respecting the sum rule exists on this hypergraph of contexts” makes sense for any orthomodular lattice, or indeed any abstract hypergraph of measurement contexts, whether or not it arises from a Hilbert space at all. Gleason’s Theorem, tied intrinsically to the operator-algebraic structure $\text{Tr}(\rho \cdot)$, does not. This asymmetry is why the contextuality literature — including the generalizations of Appendix B, and applications treating contextuality as a resource for quantum computation — is phrased combinatorially, in the language of Theorem 1.2, and essentially never in the language of Gleason’s Theorem.

APPENDIX B. ATTEMPTS BEYOND CLASSICAL PROPOSITIONAL LOGIC

The proofs of Section 3 and Section 5 work throughout with propositional, two-valued ($\{0, 1\}$ -valued) logic. Two natural ways to go beyond that setting are to keep the values binary but enrich the logic, adding bound variables and quantifiers, or to keep the propositional structure but enrich the values, allowing outcomes in $[0, 1]$ rather than $\{0, 1\}$. This appendix records what has been done in each direction.

B.1. Richer logics: quantifiers and first-order quantum logic.



- 1981

Takeuti. Gaisi Takeuti, already known for Boolean-valued analysis, publishes *Quantum Set Theory* [Tak81]. The construction mirrors the Scott–Solovay Boolean-valued models of ZFC exactly, with the complete Boolean algebra replaced by the (σ -complete) orthomodular lattice L of projections of a von Neumann algebra: a quantum universe $V^{(L)}$ is built by transfinite recursion, and quantifiers receive a literal semantics, $\llbracket (\forall x)\varphi(x) \rrbracket = \bigwedge_{x \in V^{(L)}} \llbracket \varphi(x) \rrbracket$, an infimum taken in L . Real numbers in this universe become self-adjoint operators; this is genuine first-order (indeed set-theoretic) quantification, evaluated in a non-classical, non-Boolean truth-value algebra.

- 1996

Boos. William Boos publishes *Mathematical Quantum Theory I: Random Ultrafilters as Hidden Variables* [Boo96]. Where Takeuti changed the truth-value algebra, Boos brings to quantum theory the other main tool of modern set theory, Cohen’s method of forcing. His hidden variables are *random* ultrafilters, that is, generic objects in the sense of forcing with measure algebras, and he reads quantum probabilities through this lens. The paper is a rare attempt to address the hidden-variable question, and hence the setting of Theorem 1.2, with the methods of set theory rather than those of analysis.

- 1998–2002

Isham–Butterfield. Isham and Butterfield, joined for one paper by Hamilton, publish the four-part series *A Topos Perspective on the Kochen–Specker Theorem* [IB98, BI99, HIB00, IB02]. Motivated by the objection that the orthomodular lattice is too poorly behaved (non-distributive) to serve as a logic with an ordinary semantics at all, they discard it and use instead the internal, intuitionistic, genuinely quantified higher-order logic of a topos of presheaves over the category of commuting subalgebras (“contexts”). Distributivity is recovered; truth values become context-dependent sieves in a Heyting algebra instead of elements of an orthomodular lattice. The first paper in the series reformulates Theorem 1.2 itself as the non-existence of a global element of the associated *spectral presheaf* — a direct, quantifier-rich redescription of the central result of these notes. Isham and Döring later extend this into a full topos-theoretic reformulation of physics, *A Topos Foundation for Theories of Physics* [DI08].

- 1999

Titani. Satoko Titani’s *A Lattice-Valued Set Theory* [Tit99] generalizes Takeuti’s construction from orthomodular lattices to arbitrary complete lattices as the truth-value algebra, placing quantum set theory inside a wider family alongside Heyting-valued (intuitionistic) set theories, with general completeness results.

2007–2020s

Ozawa. Masanao Ozawa turns quantum set theory from a suggestive construction into a working technical apparatus. *Transfer Principle in Quantum Set Theory* [Oza07] proves a rigorous dictionary carrying ZFC theorems about the reals over to theorems about self-adjoint operators inside $V^{(L)}$, in the spirit of the transfer principle of nonstandard analysis; *Orthomodular-Valued Models for Quantum Set Theory* [Oza09] tightens the model theory; later work reforms the semantics to recover De Morgan’s laws and extends the framework to quantum conditionals. Ozawa’s own motivation throughout is to state and prove a rigorous, quantifier-based version of Heisenberg’s uncertainty principle — exactly the kind of first-order reasoning a purely propositional logic cannot support.

B.1.1. *What about garden-variety (classical) first-order logic?* Here the honest answer has a sting in it: it was tried, directly, in the most literal sense — treat the orthogonality relation on the closed subspaces of a Hilbert space as a classical first-order structure and ask the ordinary model-theoretic questions — and the theorems of the field itself say that it provably cannot do the job.

**1963–1964**

Mackey, Piron. Before any of the above, Mackey’s axiomatic reconstruction of quantum mechanics [Mac63] and Piron’s representation theorem [Pir64] inaugurate a different and much more classical tradition: state the physical postulates (a set of “questions,” or an atomistic, orthomodular, irreducible lattice satisfying the covering law) as ordinary axioms in ordinary set-theoretic mathematics, with no non-classical logic anywhere, and *prove*, as a classical representation theorem, that any structure satisfying them is isomorphic to the subspace lattice of a generalized Hilbert space. The “quantum” content lives entirely in which structures satisfy the axioms, not in the logic used to reason about them.

1972 onward

Foulis–Randall. Foulis and Randall’s *manuals* and *test spaces* [FR72] continue this classical-metatheory tradition at a more operational level: outcomes and tests are ordinary sets, and non-Boolean structure emerges only downstream, as a theorem about which propositions can be jointly tested, not as a feature of the logic used to derive it.

1984

Goldblatt. Robert Goldblatt’s *Orthomodularity is not elementary* [Gol84] asks the model-theoretic question directly: is the class of Hilbert lattices axiomatizable by classical first-order sentences in the language of the orthogonality relation? He proves that it is not: an infinite-dimensional pre-Hilbert space and its metric completion are elementarily equivalent as first-order structures, yet only the complete one is orthomodular. Metric completeness — exactly the ingredient Theorem 1.2 and Gleason’s Theorem both quietly rely on (closedness of subspaces, convergence on $S(H)$) — is provably invisible to classical first-order logic here.

- 2001, 2007 • **Hardy, Barrett.** Hardy's *Quantum Theory from Five Reasonable Axioms* [Har01] and Barrett's *Information Processing in Generalized Probabilistic Theories* [Bar07] revive the Mackey/Piron operational program in a modern, information-theoretic idiom — again ordinary classical mathematics, with quantum theory singled out as one solution, among a classified family, of first-order-statable convexity and composition axioms.
- 2016 • **Fritz.** Tobias Fritz's *Quantum Logic is Undecidable* [Fri16], building on Slofstra's work on embeddability problems for groups, shows that the classical first-order theory of complex Hilbert lattices is undecidable already at the level of quasi-identities: there is no algorithm deciding whether an implication between equations and orthogonality relations holds in every Hilbert space.



Remark B.1. So garden-variety first-order logic was not overlooked; it was tried on the most natural target (the orthogonality relation on subspaces) and shown, by the theorems of the field itself, to be structurally too weak to pin down the class of Hilbert lattices at all (Goldblatt) and undecidable where it does apply (Fritz). This double failure — too weak to pin down the class, undecidable where it does apply — is a large part of why the two serious efforts at a genuinely quantified quantum logic (Takeuti's line and the topos line) both abandoned classical semantics rather than push harder on it: Takeuti moved the non-classicality into the truth-value algebra, Isham–Butterfield moved it into an intuitionistic internal logic, and both, not coincidentally, gained back exactly the property (extra structure that a bare classical first-order language cannot see) that Goldblatt's theorem says is missing. The Mackey/Piron/Foulis–Randall/Hardy/Barrett line, meanwhile, never tried to axiomatize Hilbert lattices in first-order logic at all: it uses classical first-order logic only as ordinary mathematics is always written, aimed at representation theorems about a differently chosen structure (a convex state space, a test space) rather than at characterizing subspace lattices up to elementary equivalence.

B.2. Richer values: $[0, 1]$ -valued and POVM-based assignments. Theorem 1.2 rules out *sharp*, two-valued, non-contextual assignments. A natural physical question is whether an analogous no-go survives once outcomes, or the physical measurements themselves, are allowed to be *unsharp*. The answer depends on which of two things is fuzzed, and pulls in opposite directions.

Unsharp measurements are modeled as follows. An *effect* on H is a self-adjoint operator E with $0 \leq E \leq I$, that is, with spectrum in $[0, 1]$. A *positive-operator-valued measure* (POVM) with finitely many outcomes is a finite family $(E_1, \dots, E_n)$ of effects with

$$E_1 + \dots + E_n = I.$$

By the Born rule, a system in the state given by a unit vector ψ yields outcome i with probability $\langle \psi | E_i | \psi \rangle$; the condition $\sum_i E_i = I$ says exactly that these probabilities sum to 1. A projective measurement, i.e. an orthogonal decomposition of I into projections as in Definition 1.1, is the special case in which every E_i is a projection. Physically, a POVM is what one obtains by coupling the system to an auxiliary system and performing an ordinary projective measurement on the combined system (Naimark's dilation theorem says every POVM arises this way). The essential new feature is that the effects of a POVM need not be mutually orthogonal, so a POVM may have more outcomes than $\dim H$, and a single effect may belong to many different POVMs.

B.2.1. Fuzzing the measurements, keeping assignments sharp: the no-go strengthens, even at $\dim H = 2$. The analogue of a two-valued quantum measure is now a map v from effects to $\{0, 1\}$ such that, for every POVM $(E_1, \dots, E_n)$, exactly one E_i has $v(E_i) = 1$: each measurement has exactly one outcome. Non-contextuality is built into the requirement that v is a function of the effect alone, not of the POVM in which it appears. In $\dim H = 2$ the projections give too few constraints for a contradiction: the only projective measurements are the pairs $\{P, I - P\}$, and a two-valued quantum measure exists (see Remark 1.4). Effects give many more constraints, because the same effect appears in many POVMs, with different companions. Cabello [Cab03] exhibited a finite family of POVMs on the qubit $\mathbb{C}^2$ whose effects admit no such v . Thus the Kochen–Specker obstruction extends past sharp projections, into a regime, $\dim H = 2$, that Theorem 1.2 does not reach.

B.2.2. Fuzzing the assignments, keeping measurements sharp: the naive no-go can be evaded, and needs a better physical definition to survive. Replace the deterministic $v(P) \in \{0, 1\}$ by a probabilistic response function $\xi_P(\lambda) \in [0, 1]$: the probability of outcome 1 given hidden ontic state λ (“ontic” is a term from philosophy, where the *ontic*, what actually exists, is distinguished from the *epistemic*, what is known about it. The intent is that λ is the complete physical state of the system, whether or not anyone knows it. In the mathematical formalism, λ is a point of a measure space Λ). Barrett and Kent [BK04] observed that once a real experiment resolves outcomes only to finite precision, non-contextual hidden-variable models built from such fuzzy response functions can reproduce quantum statistics to arbitrary accuracy — so naive basis-independence becomes physically toothless once assignments are allowed to be fuzzy. Spekkens [Spe05] closed this gap with a fully operational definition of non-contextuality (operationally indistinguishable procedures must receive identical ontological representations), under which fresh no-go theorems survive, including ones that already hold at $\dim H = 2$.

B.2.3. Extending the outcome alphabet directly. Ramanathan [Ram24] proves a Kochen–Specker-type theorem for assignments valued in a finite alphabet $\{0, p, 1 - p, 1\} \subseteq [0, 1]$ rather than only $\{0, 1\}$, for finite vector configurations in $\mathbb{C}^3$; the case $p = \frac{1}{2}$ rules out “fundamentally binary” hidden-variable theories.



Remark B.2. Fuzzing the *measurements* strengthens Theorem 1.2 (Cabello, reaching $\dim H = 2$); fuzzing the *assignments* under the naive definition of non-contextuality weakens it (Barrett–Kent), which is why Spekkens’s operational redefinition was needed to recover a robust no-go there; extending the outcome set while keeping assignments deterministic recovers a close cousin of the original theorem outright (Ramanathan). In each case the substantive work is a fresh rigidity argument playing the role Gleason’s Theorem played earlier in these notes — never a restatement of the bare fact that local assignments respect the connectives.

REFERENCES

- [Alf71] E. M. Alfsen, *Compact convex sets and boundary integrals*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 57, Springer, Berlin, 1971.
- [AOW11] M. Arends, J. Ouaknine, and C. W. Wampler, *On searching for small Kochen–Specker vector systems*, Graph-Theoretic Concepts in Computer Science (WG 2011), Lecture Notes in Computer Science, vol. 6986, Springer, 2011, pp. 23–34.
- [Bar07] J. Barrett, *Information processing in generalized probabilistic theories*, Phys. Rev. A **75** (2007), 032304.
- [Bel66] J. S. Bell, *On the problem of hidden variables in quantum mechanics*, Rev. Mod. Phys. **38** (1966), 447–452.

- [Bel87] ———, *Speakable and unspeakable in quantum mechanics: Collected papers on quantum philosophy*, Cambridge University Press, Cambridge, 1987.
- [Ber14] R. A. Bertlmann, *John Bell and the nature of the quantum world*, J. Phys. A: Math. Theor. **47** (2014), 424007.
- [BI99] J. Butterfield and C. J. Isham, *A topos perspective on the Kochen–Specker theorem: II. Conceptual aspects and classical analogues*, Int. J. Theor. Phys. **38** (1999), 827–859.
- [BK04] J. Barrett and A. Kent, *Non-contextuality, finite precision measurement and the Kochen–Specker theorem*, Stud. Hist. Phil. Mod. Phys. **35** (2004), 151–176, arXiv:quant-ph/0309017.
- [Boo96] William Boos, *Mathematical quantum theory I: Random ultrafilters as hidden variables*, Synthese **107** (1996), no. 1, 83–143.
- [Cab03] A. Cabello, *Kochen–Specker theorem for a single qubit using positive operator-valued measures*, Phys. Rev. Lett. **90** (2003), 190401, arXiv:quant-ph/0210082.
- [CEGA96] A. Cabello, J. M. Estebaranz, and G. García-Alcaine, *Bell–Kochen–Specker theorem: A proof with 18 vectors*, Phys. Lett. A **212** (1996), 183–187.
- [DI08] A. Döring and C. J. Isham, *A topos foundation for theories of physics: I–IV*, J. Math. Phys. **49** (2008), 053515–053518.
- [End01] H. B. Enderton, *A mathematical introduction to logic*, 2nd ed., Academic Press, 2001, Compactness theorem for propositional logic.
- [FR72] D. J. Foulis and C. H. Randall, *Operational statistics. I. Basic concepts*, J. Math. Phys. **13** (1972), 1667–1675, and subsequent papers on manuals and test spaces.
- [Fri16] T. Fritz, *Quantum logic is undecidable*, 2016, arXiv:1607.05870.
- [Gle57] A. M. Gleason, *Measures on the closed subspaces of a Hilbert space*, J. Math. Mech. **6** (1957), 885–893.
- [Gol84] R. Goldblatt, *Orthomodularity is not elementary*, J. Symbolic Logic **49** (1984), 401–404.
- [Ham03] J. Hamhalter, *Quantum measure theory*, Fundamental Theories of Physics, vol. 134, Kluwer Academic Publishers, Dordrecht, 2003.
- [Har01] L. Hardy, *Quantum theory from five reasonable axioms*, 2001, arXiv:quant-ph/0101012.
- [Her35] Grete Hermann, *Die naturphilosophischen Grundlagen der Quantenmechanik*, Abhandlungen der Fries’schen Schule, Neue Folge **6** (1935), 75–152.
- [HIB00] J. Hamilton, C. J. Isham, and J. Butterfield, *A topos perspective on the Kochen–Specker theorem: III. Von Neumann algebras as the base category*, Int. J. Theor. Phys. **39** (2000), 1413–1436, arXiv:quant-ph/9911020.
- [IB98] C. J. Isham and J. Butterfield, *A topos perspective on the Kochen–Specker theorem: I. Quantum states as generalised valuations*, Int. J. Theor. Phys. **37** (1998), 2669–2733, arXiv:quant-ph/9803055.
- [IB02] ———, *A topos perspective on the Kochen–Specker theorem: IV. Interval valuations*, Int. J. Theor. Phys. **41** (2002), 613–639.
- [Jam74] Max Jammer, *The philosophy of quantum mechanics*, Wiley, New York, 1974.
- [JP63] J. M. Jauch and C. Piron, *Can hidden variables be excluded in quantum mechanics?*, Helv. Phys. Acta **36** (1963), 827–837.
- [Kad51] R. V. Kadison, *Order properties of bounded self-adjoint operators*, Proc. Amer. Math. Soc. **2** (1951), 505–510.
- [Ker94] M. Kernaghan, *Bell–Kochen–Specker theorem for 20 vectors*, J. Phys. A: Math. Gen. **27** (1994), L829–L830.
- [KS65a] Simon Kochen and Ernst P. Specker, *The calculus of partial propositional functions*, Logic, Methodology and Philosophy of Science (Proceedings of the 1964 International Congress, Jerusalem) (Y. Bar-Hillel, ed.), North-Holland, Amsterdam, 1965, pp. 45–57.
- [KS65b] ———, *Logical structures arising in quantum theory*, The Theory of Models (Proceedings of the 1963 International Symposium at Berkeley) (J. W. Addison, L. Henkin, and A. Tarski, eds.), North-Holland, Amsterdam, 1965, pp. 177–189.
- [KS67] S. Kochen and E. P. Specker, *The problem of hidden variables in quantum mechanics*, J. Math. Mech. **17** (1967), 59–87.
- [LBG24] Z. Li, C. Bright, and V. Ganesh, *A SAT solver and computer algebra attack on the minimum Kochen–Specker problem*, Proc. IJCAI 2024, 2024, arXiv:2306.13319.
- [Mac57] George W. Mackey, *Quantum mechanics and Hilbert space*, Amer. Math. Monthly **64** (1957), no. 8, part II, 45–57.

- [Mac63] G. W. Mackey, *Mathematical foundations of quantum mechanics*, W. A. Benjamin, New York, 1963.
- [MC88] C. Mann and R. Crease, *Interview: John Bell*, *Omni* **10** (1988), no. 8, 84–86, 88, 90, 92, 121.
- [MS18] N. David Mermin and Rüdiger Schack, *Homer nodded: von Neumann’s surprising oversight*, *Found. Phys.* **48** (2018), 1007–1020.
- [Oza07] M. Ozawa, *Transfer principle in quantum set theory*, *J. Symbolic Logic* **72** (2007), 625–648, arXiv:math/0604349.
- [Oza09] ———, *Orthomodular-valued models for quantum set theory*, *Rev. Symbolic Logic* (2009), arXiv:0908.0367.
- [Per91] A. Peres, *Two simple proofs of the Kochen–Specker theorem*, *J. Phys. A: Math. Gen.* **24** (1991), L175–L178.
- [Per93] ———, *Quantum theory: Concepts and methods*, Kluwer, Dordrecht, 1993, Reports the unpublished 31-vector Conway–Kochen construction in $\mathbb{R}^3$.
- [Pir64] C. Piron, *Axiomatique quantique*, *Helv. Phys. Acta* **37** (1964), 439–468.
- [PMMM05] M. Pavičić, J.-P. Merlet, B. McKay, and N. D. Megill, *Kochen–Specker vectors*, *J. Phys. A: Math. Gen.* **38** (2005), no. 7, 1577–1592, arXiv:quant-ph/0409014.
- [Ram24] R. Ramanathan, *Generalised Kochen–Specker theorem for finite non-deterministic outcome assignments*, *npj Quantum Information* **10** (2024), 99.
- [RS80] Michael Reed and Barry Simon, *Methods of modern mathematical physics. I: Functional analysis*, revised and enlarged ed., Academic Press, New York, 1980.
- [Rud87] W. Rudin, *Real and complex analysis*, 3rd ed., McGraw-Hill, New York, 1987.
- [Spe60] Ernst Specker, *Die Logik nicht gleichzeitig entscheidbarer Aussagen*, *Dialectica* **14** (1960), 239–246.
- [Spe05] R. W. Spekkens, *Contextuality for preparations, transformations, and unsharp measurements*, *Phys. Rev. A* **71** (2005), 052108, arXiv:quant-ph/0406166.
- [Tak81] G. Takeuti, *Quantum set theory*, *Current Issues in Quantum Logic* (E. G. Beltrametti and B. C. van Fraassen, eds.), *Ettore Majorana International Science Series*, Plenum Press, New York, 1981, pp. 303–322.
- [Tit99] S. Titani, *A lattice-valued set theory*, *Archive for Mathematical Logic* **38** (1999), 395–421.
- [vN32] J. von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer, Berlin, 1932.